

Client Interface (CIF) Full Microarchitecture Specification

Contents

1	Overview	6
1.1	Global Assumptions	6
2	Parameters	6
3	Write Path	7
3.1	Overview	7
3.2	Block Descriptions	7
3.3	Write Packet Format	8
3.4	Write Path Backpressure Chain	8
4	Read Path	9
4.1	Overview	9
4.2	Block Descriptions	9
4.3	Read Packet Format	11
4.4	Read Path Backpressure Chain	11
5	CIF $\leftrightarrow$ MC Core Interface	12
5.1	Write Interface (CIF $\rightarrow$ MC Core)	12
5.2	Read Interface (CIF $\rightarrow$ MC Core)	12
5.3	Completion Interface (MC Core $\rightarrow$ CIF)	12
5.4	Packet Type Encoding	12
5.5	Tag Encoding	13
6	Edge Cases and Design Decisions	13
6.1	Invalid Request Handling — Write Path	13
6.2	Invalid Request Handling — Read Path	13
6.3	Protocol Violation Handling	13
6.4	Error Response Path	13
6.5	Out-of-Order Completion Handling	13
6.6	ROB Sizing and Overflow Prevention	14
6.7	Seqnum Ordering Guarantee	14
6.8	N_CLIENTS Scalability	14
7	Interface Signal Table	14

7.1	AXI Slave Interface (AXI Master → CIF)	14
7.2	Write Path: AW Metadata FIFO → Request Validator	16
7.3	Write Path: Request Validator → Tag Allocator / Error Handler	17
7.4	Write Path: Error Handler → Tag Allocator	17
7.5	Shared: Tag Allocator → Burst Splitter	17
7.6	Shared: Tag Allocator → Response Tracker / RTT	18
7.7	Shared: Burst Splitter → Packet Formation Unit (Write)	19
7.8	Write Path: W Data FIFO → Packet Formation Unit	19
7.9	Write Path: Packet Formation Unit → Write Packet FIFO	19
7.10	Write Path: Write Packet FIFO → MC Core	20
7.11	Write Path: Response Tracker → B Response Generator	20
7.12	Write Path: Response Tracker → Tag Allocator (Free)	20
7.13	Read Path: AR Metadata FIFO → Request Validator	21
7.14	Read Path: Request Validator → Tag Allocator / Error Handler	21
7.15	Read Path: Error Handler → Tag Allocator	22
7.16	Shared: Burst Splitter → Packet Formation Unit (Read)	22
7.17	Read Path: Packet Formation Unit → Read Packet FIFO	23
7.18	Read Path: Read Packet FIFO → MC Core	23
7.19	MC Core Completion Interface (MC Core → CIF)	23
7.20	Read Path: ROB → Merge Logic	24
7.21	Read Path: RTT → ROB / Merge Logic	24
7.22	Read Path: Merge Logic → R Response Generator	24
7.23	ROB Backpressure Control	25
8	Arbitration Policy	26
8.1	Overview	26
8.2	Transaction Tag Allocator	26
8.3	Burst Splitter	27
9	Reset and Initialization Sequence	28
9.1	Overview	28
9.2	Reset Signal	28
9.3	Power-On Reset Sequence	28
9.4	Soft Reset Sequence	29
9.5	BRAM Initialization Note	29
10	Error Propagation Timing	31
10.1	Overview	31
10.2	Write Path Error Propagation	32
10.3	Read Path Error Propagation	33
10.4	Split Transaction Error Propagation	33
10.5	Err_write W Beat Handling	34
10.6	Timing Summary	34
A	Burst Splitter	34
A.1	Overview	34
A.2	Stage 1 — Decode & Compute (1 cycle, registered output)	34
A.3	Burst Type Decoder	35

A.4	Row Boundary Checker	35
A.5	WRAP Address Generator	35
A.6	Split Point Calc	35
A.7	Beat Count Calc	35
A.8	S1→S2 Pipeline Register (clocked)	35
A.9	Stage 2 — Emit & Track	35
A.10	Emit FSM	35
A.11	Sub-transaction A Mux	36
A.12	Sub-transaction B Mux	36
A.13	Beat Tracker	36
A.14	WRAP Beat Address Table	36
A.15	Pkt Assembler	36
A.16	Backpressure & Control Signals	36
A.17	Three Paths Through the Splitter	36
A.18	Internal State	37
A.19	Timing Notes	37
B	Request Validator	37
B.1	Overview	37
B.2	Design Parameters	38
B.3	Internal Blocks	38
B.4	Address Range Check	38
B.5	Output Mux	38
B.6	Interface Signals	38
B.7	Inputs	38
B.8	Outputs	39
B.9	Functional Behaviour	39
B.10	Timing Notes	39
B.11	Edge Cases	39
C	Transaction Tag Allocator	40
C.1	Overview	40
C.2	Design Parameters	40
C.3	Internal Blocks	40
C.4	Occupancy Counter Array	40
C.5	Seqnum Counter Array	41
C.6	Stall Logic	41
C.7	Allocate Logic	41
C.8	Free Logic	41
C.9	Interface Signals	42
C.10	Inputs	42
C.11	Outputs	42
C.12	Functional Behaviour	42
C.13	Allocation Flow	42
C.14	Stall Flow	43
C.15	Free Flow	43
C.16	Timing Notes	43
C.17	Edge Cases	43

D	Response Tracker	44
D.1	Overview	44
D.2	Design Parameters	44
D.3	Internal Blocks	44
D.4	Register File	44
D.5	Alloc Write Logic	45
D.6	Completion Update Logic	45
D.7	Compare Logic	45
D.8	Head Pointer Array	46
D.9	Emit Logic	46
D.10	Free Logic	46
D.11	Interface Signals	46
D.12	Inputs	46
D.13	Outputs	47
D.14	Internal Signals	47
D.15	Functional Behaviour	48
D.16	Allocation Flow	48
D.17	Completion Flow	48
D.18	Emit Flow	48
D.19	Free Flow	48
D.20	Timing Notes	49
D.21	Edge Cases	49
E	Read Transaction Table (RTT)	49
E.1	Overview	49
E.2	Design Parameters	50
E.3	Internal Blocks	50
E.4	RTT Register File	50
E.5	Alloc Write Logic	50
E.6	Completion Update Logic	51
E.7	Compare Logic	51
E.8	Free Logic	51
E.9	Interface Signals	51
E.10	Inputs	51
E.11	Outputs	52
E.12	Internal Signal Table	52
E.13	Functional Behaviour	52
E.14	Allocation Flow	52
E.15	Completion Flow	52
E.16	Free Flow	53
E.17	rob_last Routing	53
E.18	Timing Notes	53
E.19	Edge Cases	53
F	Reorder Buffer (ROB)	54
F.1	Overview	54
F.2	Blocks	55
F.3	Free List	55

F.4	Occupancy Counter	55
F.5	Lookup Table	55
F.6	ROB BRAM	55
F.7	Completion Input	55
F.8	Head Pointer Array	55
F.9	Emit Logic	56
F.10	PKT_READY Control	56
F.11	Allocation Path	56
F.12	Completion Path	56
F.13	Emit Path	57
F.14	Backpressure & Overflow Prevention	57
F.15	Key Parameters	57
F.16	Timing Notes	58

1 Overview

The Controller Interface (CIF) is the AXI-facing front-end of the memory subsystem. It translates AXI4 read and write transactions into normalized, fixed-width packets suitable for MC Core processing. The CIF handles burst splitting, transaction tagging, packet formation, response tracking, and in-order AXI response generation. MC Core remains transaction-agnostic and operates purely on a per-packet basis.

1.1 Global Assumptions

- AXI4 protocol.
- Data width: 512 bits (64 bytes). Parameterizable via RTL.
- Address width: 32 bits (FPGA initial configuration). Parameterizable.
- AXID width: 6 bits (64 unique AXI IDs).
- Supported burst types: INCR and WRAP. FIXED not supported.
- Multiple outstanding transactions supported per AXID (up to 4).
- Maximum burst length: 256 beats (AXI4 maximum).
- Narrow INCR and unaligned INCR transfers supported.
- Narrow WRAP and unaligned WRAP transfers not supported.
- Out-of-order (OoO) completions from MC Core are supported and resolved within CIF.
- Protocol violations (illegal burst type, unaligned WRAP, etc.) are design-time assertions only. Runtime validation is limited to address legality checks.

2 Parameters

Parameter	Default	Description
AXID_W	6	AXI ID width. Supports up to $2^6 = 64$ unique IDs.
ADDR_W	32	Address width in bits.
DATA_W	512	Data path width in bits.
N_AR	128	AR Metadata FIFO depth ($= 4 \times \text{N_CLIENTS}$).
N_AW	128	AW Metadata FIFO depth ($= 4 \times \text{N_CLIENTS}$).
N_W	64	W Data FIFO depth.
N_PKT	64	Packet FIFO depth (read and write).
MC_RD_BUF	64	MC Core internal read buffer depth. Defines ROB depth.
ROB_WATERMARK	26	MC-side watermark for backpressure signalling.
MAX_OUTSTANDING	4	Maximum outstanding transactions per AXID.
N_CLIENTS	32	AXI client count. Fixed at compile time (max 32).

All parameters are configurable at synthesis time.

Table 1: CIF Global Parameters

Note on N_CLIENTS. The number of AXI clients is fixed at synthesis time. Each client is statically allocated `MAX_OUTSTANDING = 4` slots in the AR and AW metadata FIFOs, giving `N_AR = N_AW = 4 × N_CLIENTS`. Idle clients retain their allocated slots; dynamic reallocation across clients is deferred to a future CIF revision.

3 Write Path

3.1 Overview

The write path accepts AXI write transactions on the AW and W channels, validates them, splits bursts into normalized 64-byte packets, and forwards them to MC Core. Per-packet completions are tracked and aggregated to generate a single AXI B channel response per transaction.

3.2 Block Descriptions

AW Metadata FIFO (depth = $N_AW = 4 \times N_CLIENTS$). Captures and buffers incoming AW channel metadata: AWADDR, AWLEN, AWSIZE, AWBURST, AWID. Decouples AXI request arrival from downstream validation and scheduling. Each AXI client is statically allocated 4 slots. AWREADY is driven low when no slots are available for the requesting client.

W Data FIFO (depth = N_W). Captures and buffers AXI write data beats: WDATA, WSTRB, WLAST. Operates independently of the AW channel. W beats flow in freely regardless of AW validation status.

Request Validator. Pops entries from the AW Metadata FIFO and performs address legality checks combinationally:

- Address range validation.

Protocol-level constraints (burst type, WRAP legality, alignment, burst length) are enforced as design-time assertions and are not checked at runtime — compliant AXI masters are assumed. Valid requests are forwarded to the Transaction Tag Allocator. Address errors are forwarded to the Error Handler along with AWID.

Transaction Tag Allocator. Shared across all $N_CLIENTS$ AXI clients. Allocates a unique `txn_tag` per validated transaction:

$txn_tag[7:0] = AXID[5:0] + seqnum[1:0]$
seqnum: monotonically increasing per AXID, wraps at 4

Accepts inputs from both the Request Validator (valid requests) and the Error Handler (NOP transactions). Seqnum allocation is blocked when all 4 outstanding slots for an AXID are occupied. Tags are freed when the Response Tracker emits a BRESP for the corresponding transaction.

Burst Splitter (two-stage pipeline). Shared across all $N_CLIENTS$ AXI clients. Converts validated AXI burst requests into normalized beat-level packets.

- *Stage 1 — Row Boundary Split:* Splits bursts such that no sub-transaction crosses a DRAM row boundary. Generates row-confined sub-transactions.
- *Stage 2 — Command Alignment & Packetization:* Aligns packets to memory-command boundaries. Generates byte-enable wrapper masks for partial or edge-aligned packets.

Internal state: `base_addr`, `beat_idx`, `beats_left`, `burst_type`, `burst_len`, `txn_tag`.

Packet Formation Unit. Assembles normalized 618-bit write packets:

<code>pkt_type</code>	<code>[1:0]</code>	2 bits	(00=Write , 01=Read , 10=Err_write , 11=Err_read)
<code>txn_tag</code>	<code>[9:2]</code>	8 bits	
<code>addr</code>	<code>[41:10]</code>	32 bits	

data [553:42] 512 bits
 wrapper [617:554] 64 bits (1 bit = 1 byte validity)
 Total: 618 bits

One AXI beat maps directly to one normalized packet. No beat accumulation required.

Packet FIFO (depth = N_PKT). Buffers normalized write packets before transfer to MC Core. Provides decoupling between CIF packet generation rate and MC Core processing rate.

Response Tracker. Shared across all N_CLIENTS AXI clients. Tracks outstanding write transactions on a per-tag basis. Stores: `txn_tag`, `AXID` (recovered as `txn_tag[7:2]`), `expected_packets` (from `AWLEN+1`), `completed_packets`, `status`. Emits a single `BRESP` when:

`completed_packets == expected_packets`

On receiving an error status from MC Core (corresponding to an `Err_write` packet), passes `DECERR` to the B Response Generator.

B Response Generator. Drives the AXI B channel: `BVALID`, `BRESP`, `BID`. `BID` is recovered from `txn_tag[7:2]`. Generates `DECERR` when an error status is received from the Response Tracker, and `OKAY` otherwise. Single path to the AXI B channel — no arbitration mux required.

Error Handler. Receives address error notifications from the Request Validator. Sets `pkt_type = Err_write` and forwards the transaction into the normal pipeline via the Transaction Tag Allocator. W beats for the transaction flow through the pipeline without draining or discarding. MC Core receives the `Err_write` packet and returns an error completion; `DECERR` is subsequently generated by the B Response Generator on the AXI B channel.

3.3 Write Packet Format

Field	Width (bits)	Description
<code>pkt_type</code>	2	Packet type: 00=Write, 01=Read, 10=Err_write, 11=Err_read
<code>txn_tag</code>	8	Transaction tag (<code>AXID[5:0] + seqnum[1:0]</code>)
<code>addr</code>	32	Write address
<code>data</code>	512	Write data payload
<code>wrapper</code>	64	Byte-level validity mask (1 bit = 1 byte)
Total	618	<code>PKT_DATA[617:0]</code>

Table 2: Write Packet Format

3.4 Write Path Backpressure Chain

Backpressure propagates right-to-left through the write pipeline:

MC Core busy $\rightarrow$ `PKT_READY = 0`

- $\rightarrow$ Packet FIFO fills up
- $\rightarrow$ Packet Formation Unit stalls
- $\rightarrow$ Burst Splitter stalls
- $\rightarrow$ W Data FIFO absorbs backpressure
- $\rightarrow$ W Data FIFO full $\rightarrow$ `WREADY = 0`

$\rightarrow$ AW Metadata FIFO full $\rightarrow$ AWREADY = 0

This avoids materializing large bursts upfront and keeps buffering requirements bounded.

4 Read Path

4.1 Overview

The read path accepts AXI read transactions on the AR channel, validates them, splits bursts into normalized read request packets, forwards them to MC Core, and reassembles in-order read data for AXI R channel delivery. MC Core may return completions out of order; the CIF Reorder Buffer restores per-AXID AXI ordering before response generation.

4.2 Block Descriptions

AR Metadata FIFO (depth = N_AR = 4 × N_CLIENTS). Captures and buffers incoming AR channel metadata: ARADDR, ARLEN, ARSIZE, ARBURST, ARID. Decouples AXI request arrival from downstream processing. Each AXI client is statically allocated 4 slots. ARREADY is driven low when no slots are available for the requesting client.

Request Validator. Performs address legality checks only. Protocol-level constraints are enforced as design-time assertions and are not checked at runtime. Valid requests proceed to the Transaction Tag Allocator. Address errors are forwarded to the Error Handler with ARID preserved.

Transaction Tag Allocator. Shared across all N_CLIENTS AXI clients. Same encoding as the write path:

$$\text{txn_tag}[7:0] = \text{AXID}[5:0] + \text{seqnum}[1:0]$$

Accepts inputs from both the Request Validator (valid requests) and the Error Handler (NOP transactions). Seqnum is monotonically increasing per AXID and wraps at 4.

Burst Splitter. Shared across all N_CLIENTS AXI clients. Same two-stage pipeline as write path. Generates per-beat read request contexts containing `txn_tag` and `addr`.

Packet Formation Unit. Assembles normalized 42-bit read request packets:

<code>pkt_type</code>	[1:0]	2 bits	(00=Write, 01=Read, 10=Err_write, 11=Err_read)
<code>txn_tag</code>	[9:2]	8 bits	
<code>addr</code>	[41:10]	32 bits	
Total:	42 bits		PKT_DATA[41:0]

Packet FIFO (depth = N_PKT). Buffers normalized read request packets before forwarding to MC Core.

Read Transaction Table (RTT). Allocated at tag assignment time. Stores per-transaction metadata:

- AXID
- `expected_packets` (from ARLEN+1)
- `completed_packets`

Asserts `rob_last` when `completed_packets == expected_packets`, signalling the final beat of a transaction to Merge Logic. Entry is freed after `rob_last` is emitted.

Reorder Buffer (ROB). Stores out-of-order read completions from MC Core and emits them in-order per AXID. Depth is tied to `MC_RD_BUF` — the ROB never needs more slots than MC Core can have in-flight simultaneously.

Structure:

ROB[ROBDEPTH] — flat buffer, depth = 256

Each slot:

valid 1 bit
data 512 bits
status 4 bits

Free list: 256 entries

Lookup table: (AXID, seqnum) → slot index

On tag allocation:

pop free slot index from free list

store mapping: (AXID, seqnum) → slot index in lookup table

On MC completion:

extract AXID = tag[7:2], seqnum = tag[1:0]

lookup slot index from (AXID, seqnum)

ROB[slot].data = completion data

ROB[slot].status = completion status

ROB[slot].valid = 1

emit loop:

```
while ROB[slot_at_head].valid == 1:
    emit to Merge Logic
    ROB[slot_at_head].valid = 0
    push slot back to free list
    advance head pointer for AXID
```

Depth: `ROB_DEPTH` = 256. Implemented as dual-port SRAM/BRAM. Watermark threshold: `ROB_DEPTH` - `ROB_WATERMARK` = 256 - 128 = 128.

Merge Logic. Merge Logic is a purely combinational block that sits between the ROB and the R Response Generator. It has no internal storage and introduces no buffering. Its responsibilities are:

- **RID recovery:** Extracts `RID[5:0] = AXID[5:0]` from the ROB-emitted beat. Per-AXID ordering is already guaranteed by the ROB head pointer scheme — Merge Logic does not perform any reordering.
- **RLAST generation:** Asserts `RLAST` when `rob_last` is received coincident with the final data beat. `rob_last` arrives as a sideband signal from the RTT in parallel with the ROB data beat — Merge Logic combines both in the same cycle.
- **Status passthrough:** Forwards `status[3:0]` from the ROB to the R Response Generator unchanged. The R Response Generator maps non-zero status to `DECERR` on the AXI R channel.

- **Data passthrough:** Forwards `data[511:0]` from the ROB to the R Response Generator unchanged.

Merge Logic does not arbitrate between AXIDs — the ROB Emit Logic’s priority encoder already selects one AXID per cycle. Merge Logic sees a single beat per cycle at most and forwards it directly. No inter-AXID buffering or sequencing is required here.

R Response Generator. AXI R channel driver. Restores RID from `txn_tag[7:2]`, generates `RLAST`, `RRESP`, asserts `RVALID`, and drives `RDATA`. Generates `DECERR` when an error status is received from the ROB, and `OKAY` otherwise. Single path to the AXI R channel — no arbitration mux required.

Error Handler. Receives address error notifications from the Request Validator with `ARID` preserved. Sets `pkt_type = Err_read` and forwards the transaction into the normal pipeline via the Transaction Tag Allocator. MC Core receives the `Err_read` packet and returns an error completion; `DECERR` is subsequently generated by the R Response Generator on the AXI R channel.

4.3 Read Packet Format

Field	Width (bits)	Description
<code>pkt_type</code>	2	Packet type: 00=Write, 01=Read, 10=Err_write, 11=Err_read
<code>txn_tag</code>	8	Transaction tag (<code>AXID[5:0] + seqnum[1:0]</code>)
<code>addr</code>	32	Read address
Total	42	PKT_DATA[41:0]

Table 3: Read Packet Format

4.4 Read Path Backpressure Chain

MC Core does not support backpressure on the completion interface (`RESP_READY` must always be 1). CIF must therefore ensure the ROB never fills completely. Backpressure is applied by halting new read requests to MC Core when the ROB occupancy reaches the watermark threshold:

ROB occupancy $\geq$ ROB_DEPTH − ROB_WATERMARK (= 256 − 128 = 128)

- `PKT_READY = 0` (stop sending new requests to MC)
- Packet FIFO fills up
- Burst Splitter stalls
- AR Metadata FIFO absorbs backpressure
- AR Metadata FIFO full → `ARREADY = 0`

The `ROB_WATERMARK = 128` headroom ensures all in-flight requests already accepted by MC Core can return completions without overflowing the ROB. This value is derived conservatively from MC Core’s confirmed maximum pipeline depth of 100, rounded up to account for scheduler speculation and forward pressure.

Signal	Width	Description
PKT_VALID	1	Packet valid
PKT_READY	1	MC Core ready to accept
PKT_DATA	618	Serialized write packet

Table 4: Write Request Interface

5 CIF ↔ MC Core Interface

5.1 Write Interface (CIF → MC Core)

5.2 Read Interface (CIF → MC Core)

Signal	Width	Description
PKT_VALID	1	Packet valid
PKT_READY	1	MC Core ready to accept
PKT_DATA	42	Serialized read request packet

Table 5: Read Request Interface

5.3 Completion Interface (MC Core → CIF)

Agreed with MC Core team. RESP_READY must be held high at all times by CIF — MC Core does not support backpressure on the completion path.

Signal	Width	Description
RESP_VALID	1	Completion valid
RESP_READY	1	Must always be 1 (CIF always ready)
resp_type	2	00=RD_DATA, 01=WR_ACK, 10=ERR
tag	8	tag[7:2]=AXID, tag[1:0]=seqnum
data	512	Read data (valid for RD_DATA only)
status	4	Status/error code (encoding TBD with MC team)

Table 6: MC Core Completion Interface — agreed with MC Core team

5.4 Packet Type Encoding

The `pkt_type` field is carried in bits [1:0] of all CIF-to-MC-Core packets. Agreed with MC Core team.

`pkt_type`[1:0]:

- 00 = Write — normal write transaction
- 01 = Read — normal read transaction
- 10 = Err_write — address-error write; MC Core returns error completion
- 11 = Err_read — address-error read; MC Core returns error completion

MC Core discards data for `Err_write` and `Err_read` packets and returns an error status on the completion interface. CIF maps this to `DECERR` on the AXI B or R channel respectively.

5.5 Tag Encoding

`txn_tag[7:0]`:
 `[7:2]` = AXID (6 bits, up to 64 unique IDs)
 `[1:0]` = seqnum (2 bits, 0–3, per-AXID outstanding slot)

Agreed with MC Core team. MC Core returns the same `tag` value on completion that CIF embedded in the original request packet.

6 Edge Cases and Design Decisions

6.1 Invalid Request Handling — Write Path

The Request Validator performs address legality checks only at runtime. On an address error, the Error Handler sets `pkt_type` = `Err_write` and injects the transaction into the normal pipeline via the Transaction Tag Allocator. W beats for the transaction flow through the W Data FIFO and Packet Formation Unit without modification. MC Core receives the `Err_write` packet, discards the data, and returns an error completion. The B Response Generator drives `BRESP` = `DECERR` on the AXI B channel. No W beat draining or direct error bypass path is required.

6.2 Invalid Request Handling — Read Path

On an address error, the Error Handler sets `pkt_type` = `Err_read` and injects the transaction into the normal pipeline. MC Core receives the `Err_read` packet and returns an error completion. The R Response Generator drives `RRESP` = `DECERR` on the AXI R channel. No ROB slot is bypassed and no special read-path handling is required.

6.3 Protocol Violation Handling

Burst type violations, WRAP illegality, alignment errors, and maximum burst length violations are not checked at runtime. Compliant AXI4 masters are assumed. These constraints are enforced as design-time assertions in the verification environment only.

6.4 Error Response Path

Both read and write paths use a single response path — the normal Response Generator drives the AXI R and B channels respectively. The arbitration mux previously used to multiplex error and normal responses has been removed. All responses, including `DECERR`, are generated by the Response Generator based on the completion status received from MC Core.

6.5 Out-of-Order Completion Handling

MC Core may return read completions out of order. The ROB absorbs OoO completions using a flat buffer indexed by dynamically allocated slot. On completion, CIF resolves the slot index via a lookup table keyed on (`AXID`, `seqnum`). In-order emission per AXID is maintained by tracking a head pointer per AXID; a slot is emitted and freed only when the head slot for that AXID is valid. This guarantees AXI R channel ordering compliance.

6.6 ROB Sizing and Overflow Prevention

The ROB depth is sized to ensure the ROB never overflows given MC Core’s maximum in-flight pipeline depth. `MC_RD_BUF` = 100 is MC Core’s confirmed maximum pipeline depth. `ROB_DEPTH` = 256 is chosen as the next power of two above $2 \times \text{MC_RD_BUF}$, providing margin for scheduler speculation and forward pressure. Since `RESP_READY` must always be 1, CIF cannot stall incoming completions. Instead, CIF halts new read request issuance (`PKT_READY` = 0) when ROB occupancy reaches `ROB_DEPTH` - `ROB_WATERMARK` = 128. The `ROB_WATERMARK` = 128 headroom absorbs all completions in-flight within MC Core without ROB overflow.

6.7 Seqnum Ordering Guarantee

The Transaction Tag Allocator guarantees monotonically increasing seqnums per AXID. This property underpins the correctness of the ROB head-pointer ordering scheme. Seqnum allocation stalls if all 4 outstanding slots for an AXID are occupied, providing implicit backpressure on the request path.

6.8 N_CLIENTS Scalability

`N_CLIENTS` is a compile-time parameter. AR and AW metadata FIFO depths scale as $4 \times \text{N_CLIENTS}$, ensuring each client always has 4 guaranteed outstanding slots. Idle clients retain their allocated slots; unutilized capacity is not reclaimed dynamically in this revision. A future CIF revision will introduce a shared slot pool — with 2 guaranteed slots per client and the remaining (`total_depth` - $2 \times \text{N_CLIENTS}$) slots dynamically allocated — controlled by a QoS-aware arbitration fabric.

7 Interface Signal Table

This section enumerates every signal crossing a block boundary within the CIF. Signals are grouped by boundary in pipeline order: AXI slave interface first, then write path, then read path, then shared blocks, then the CIF–MC Core interface.

Handshake types used below:

- **VALID/READY** — standard AXI4 handshake; transfer occurs when both asserted.
- **VALID/READY (tied)** — `READY` permanently high; transfer occurs on `VALID`.
- **Pulse** — single-cycle strobe, no handshake.
- **Level** — combinational or registered level signal, no handshake.
- **FIFO push/pop** — implicit handshake via FIFO full/empty flags.

7.1 AXI Slave Interface (AXI Master → CIF)

Signal	W	Dir	From	To	Handshake	Description
AWID[5:0]	6	in	AXI Master	AW Meta FIFO	VALID/READY	Write transaction ID
AWADDR[31:0]	32	in	AXI Master	AW Meta FIFO	VALID/READY	Write address
AWLEN[7:0]	8	in	AXI Master	AW Meta FIFO	VALID/READY	Burst length minus 1
AWSIZE[2:0]	3	in	AXI Master	AW Meta FIFO	VALID/READY	Transfer size
AWBURST[1:0]	2	in	AXI Master	AW Meta FIFO	VALID/READY	Burst type (INCR/WRAP)
AWVALID	1	in	AXI Master	AW Meta FIFO	VALID/READY	AW channel valid
AWREADY	1	out	AW Meta FIFO	AXI Master	VALID/READY	AW channel ready; deasserted when client slot full
WDATA[511:0]	512	in	AXI Master	W Data FIFO	VALID/READY	Write data
WSTRB[63:0]	64	in	AXI Master	W Data FIFO	VALID/READY	Write byte strobes
WLAST	1	in	AXI Master	W Data FIFO	VALID/READY	Last beat of burst
WVALID	1	in	AXI Master	W Data FIFO	VALID/READY	W channel valid
WREADY	1	out	W Data FIFO	AXI Master	VALID/READY	W channel ready; deasserted when FIFO full
BVALID	1	out	B Resp Gen	AXI Master	VALID/READY	B channel valid
BREADY	1	in	AXI Master	B Resp Gen	VALID/READY	B channel ready
BRESP[1:0]	2	out	B Resp Gen	AXI Master	VALID/READY	Write response (OKAY/DECERR)
BID[5:0]	6	out	B Resp Gen	AXI Master	VALID/READY	Write response ID
ARID[5:0]	6	in	AXI Master	AR Meta FIFO	VALID/READY	Read transaction ID

ARADDR[31:0]	32	in	AXI Master	AR Meta FIFO	VALID/READY	Read address
ARLEN[7:0]	8	in	AXI Master	AR Meta FIFO	VALID/READY	Burst length minus 1
ARSIZE[2:0]	3	in	AXI Master	AR Meta FIFO	VALID/READY	Transfer size
ARBURST[1:0]	2	in	AXI Master	AR Meta FIFO	VALID/READY	Burst type (INCR/WRAP)
ARVALID	1	in	AXI Master	AR Meta FIFO	VALID/READY	AR channel valid
ARREADY	1	out	AR Meta FIFO	AXI Master	VALID/READY	AR channel ready; deasserted when client slot full
RVALID	1	out	R Resp Gen	AXI Master	VALID/READY	R channel valid
RREADY	1	in	AXI Master	R Resp Gen	VALID/READY	R channel ready
RDATA[511:0]	512	out	R Resp Gen	AXI Master	VALID/READY	Read data
RRESP[1:0]	2	out	R Resp Gen	AXI Master	VALID/READY	Read response (OKAY/DECERR)
RID[5:0]	6	out	R Resp Gen	AXI Master	VALID/READY	Read response ID
RLAST	1	out	R Resp Gen	AXI Master	VALID/READY	Last beat of read burst

7.2 Write Path: AW Metadata FIFO → Request Validator

Signal	W	Dir	From	To	Handshake	Description
AWADDR[31:0]	32	out	AW Meta FIFO	Req Validator	FIFO pop	Write address from FIFO head
AWID[5:0]	6	out	AW Meta FIFO	Req Validator	FIFO pop	AXI ID from FIFO head
AWLEN[7:0]	8	out	AW Meta FIFO	Req Validator	FIFO pop	Burst length from FIFO head
AWSIZE[2:0]	3	out	AW Meta FIFO	Req Validator	FIFO pop	Transfer size from FIFO head
AWBURST[1:0]	2	out	AW Meta FIFO	Req Validator	FIFO pop	Burst type from FIFO head

STALL	1	in	Tag Allocator	Req Validator	Level	Per-AXID stall; Req Validator holds pop when asserted
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7.3 Write Path: Request Validator → Tag Allocator / Error Handler

Signal	W	Dir	From	To	Handshake	Description
REQ_VALID	1	out	Req Validator	Tag Allocator	Pulse	Valid transaction; address check passed
AXID[5:0]	6	out	Req Validator	Tag Allocator	Pulse	AXI ID of valid request
AWLEN[7:0]	8	out	Req Validator	Tag Allocator	Pulse	Burst length of valid request
ERR_VALID	1	out	Req Validator	Error Handler	Pulse	Address error notification
AXID[5:0]	6	out	Req Validator	Error Handler	Pulse	AXI ID preserved for NOP tag allocation

7.4 Write Path: Error Handler → Tag Allocator

Signal	W	Dir	From	To	Handshake	Description
REQ_VALID	1	out	Error Handler	Tag Allocator	Pulse	NOP allocation request; <code>pkt_type</code> pre-set to <code>Err_write</code>
AXID[5:0]	6	out	Error Handler	Tag Allocator	Pulse	AXI ID for NOP tag allocation
AWLEN[7:0]	8	out	Error Handler	Tag Allocator	Pulse	Burst length forwarded from AW Metadata FIFO; required so Tag Allocator can initialise <code>exp_pkts = AWLEN+1</code> in the Response Tracker

7.5 Shared: Tag Allocator → Burst Splitter

Signal	W	Dir	From	To	Handshake	Description
TXN_TAG[7:0]	8	out	Tag Allocator	Burst Splitter	Pulse	Allocated tag; [7:2]=AXID, [1:0]=seqnum
TAG_VALID	1	out	Tag Allocator	Burst Splitter	Pulse	Tag output valid; registered, valid cycle after alloc_en
STALL	1	out	Tag Allocator	Req Validator / Error Handler	Level	Per-AXID stall; asserted when occupancy[AXID]==4

7.6 Shared: Tag Allocator → Response Tracker / RTT

Signal	W	Dir	From	To	Handshake	Description
TXN_TAG[7:0]	8	out	Tag Allocator	Resp Tracker	Pulse	Allocated write tag; indexes Response Tracker register file
AWLEN[7:0]	8	out	Tag Allocator	Resp Tracker	Pulse	Write burst length; exp_pkts = AWLEN+1
TAG_VALID	1	out	Tag Allocator	Resp Tracker	Pulse	Allocation strobe to Response Tracker
TXN_TAG[7:0]	8	out	Tag Allocator	RTT	Pulse	Allocated read tag; indexes RTT register file
ARLEN[7:0]	8	out	Tag Allocator	RTT	Pulse	Read burst length; exp_pkts = ARLEN+1
TAG_VALID	1	out	Tag Allocator	RTT	Pulse	Allocation strobe to RTT
alloc_req	1	out	Tag Allocator	ROB	Pulse	ROB slot allocation request, coincident with read TAG_VALID
AXID[5:0]	6	out	Tag Allocator	ROB	Pulse	AXI ID for ROB slot allocation

seqnum[1:0]	2	out	Tag Allocator	ROB	Pulse	Seqnum for ROB lookup table write
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7.7 Shared: Burst Splitter → Packet Formation Unit (Write)

Signal	W	Dir	From	To	Handshake	Description
pkt_valid	1	out	Burst Splitter	Write PFU	VALID/READY	Normalized beat context valid
pkt_ready	1	in	Write PFU	Burst Splitter	VALID/READY	Downstream ready; holds Emit FSM when low
addr[31:0]	32	out	Burst Splitter	Write PFU	VALID/READY	Per-beat write address
TXN.TAG[7:0]	8	out	Burst Splitter	Write PFU	VALID/READY	Transaction tag
pkt_type[1:0]	2	out	Burst Splitter	Write PFU	VALID/READY	00=Write, 10=Err_write; carried unchanged from Stage 1
last_beat	1	out	Burst Splitter	Write PFU	VALID/READY	Final beat of sub-transaction
s1_busy	1	out	Burst Splitter	AW Meta FIFO	Level	Stalls Stage 1 FIFO pop during EMIT B state

7.8 Write Path: W Data FIFO → Packet Formation Unit

Signal	W	Dir	From	To	Handshake	Description
WDATA[511:0]	512	out	W Data FIFO	Write PFU	FIFO pop	Write data beat
WSTRB[63:0]	64	out	W Data FIFO	Write PFU	FIFO pop	Byte enable strobes
WLAST	1	out	W Data FIFO	Write PFU	FIFO pop	Last beat flag

7.9 Write Path: Packet Formation Unit → Write Packet FIFO

Signal	W	Dir	From	To	Handshake	Description
PKT_DATA[617:0]	618	out	Write PFU	Write Pkt FIFO	FIFO push	Serialized 618-bit write packet: {wrapper, data, addr, txn_tag, pkt_type}
PKT_VALID	1	out	Write PFU	Write Pkt FIFO	FIFO push	Packet push strobe

7.10 Write Path: Write Packet FIFO → MC Core

Signal	W	Dir	From	To	Handshake	Description
PKT_DATA[617:0]	618	out	Write Pkt FIFO	MC Core	VALID/READY	Serialized write packet
PKT_VALID	1	out	Write Pkt FIFO	MC Core	VALID/READY	Packet valid
PKT_READY	1	in	MC Core	Write Pkt FIFO	VALID/READY	MC Core ready to accept; backpressure prop- agates upstream

7.11 Write Path: Response Tracker → B Response Generator

Signal	W	Dir	From	To	Handshake	Description
BRESP_VALID	1	out	Resp Tracker	B Resp Gen	Pulse	BRESP ready to emit; asserted cycle after txn_done evaluated
BID[5:0]	6	out	Resp Tracker	B Resp Gen	Pulse	AXI ID recovered from head_tag[7:2]
BRESP[1:0]	2	out	Resp Tracker	B Resp Gen	Pulse	OKAY if err_sticky==0; DECERR if err_sticky==1

7.12 Write Path: Response Tracker → Tag Allocator (Free)

Signal	W	Dir	From	To	Handshake	Description
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WR_FREE_VALID	1	out	Resp Tracker	Tag Allocator	Pulse	Write slot free strobe; coincident with BRESP emission
WR_FREE_TAG[7:0]	8	out	Resp Tracker	Tag Allocator	Pulse	Tag being freed; [7:2]=AXID for occupancy decrement

7.13 Read Path: AR Metadata FIFO → Request Validator

Signal	W	Dir	From	To	Handshake	Description
ARADDR[31:0]	32	out	AR Meta FIFO	Req Validator	FIFO pop	Read address from FIFO head
ARID[5:0]	6	out	AR Meta FIFO	Req Validator	FIFO pop	AXI ID from FIFO head
ARLEN[7:0]	8	out	AR Meta FIFO	Req Validator	FIFO pop	Burst length from FIFO head
ARSIZE[2:0]	3	out	AR Meta FIFO	Req Validator	FIFO pop	Transfer size from FIFO head
ARBURST[1:0]	2	out	AR Meta FIFO	Req Validator	FIFO pop	Burst type from FIFO head
STALL	1	in	Tag Allocator	Req Validator	Level	Per-AXID stall; Req Validator holds pop when asserted

7.14 Read Path: Request Validator → Tag Allocator / Error Handler

Signal	W	Dir	From	To	Handshake	Description
REQ_VALID	1	out	Req Validator	Tag Allocator	Pulse	Valid read request; address check passed
AXID[5:0]	6	out	Req Validator	Tag Allocator	Pulse	AXI ID of valid read request
ARLEN[7:0]	8	out	Req Validator	Tag Allocator	Pulse	Burst length of valid read request
ERR_VALID	1	out	Req Validator	Error Handler	Pulse	Address error notification

AXID[5:0]	6	out	Req Validator	Error Handler	Pulse	AXI ID preserved for NOP tag allocation
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7.15 Read Path: Error Handler → Tag Allocator

Signal	W	Dir	From	To	Handshake	Description
REQ_VALID	1	out	Error Handler	Tag Allocator	Pulse	NOP allocation request; <code>pkt_type</code> pre-set to <code>Err_read</code>
AXID[5:0]	6	out	Error Handler	Tag Allocator	Pulse	AXI ID for NOP tag allocation
ARLEN[7:0]	8	out	Error Handler	Tag Allocator	Pulse	Burst length forwarded from AR Metadata FIFO; required so Tag Allocator can initialise <code>exp_pkts = ARLEN+1</code> in the RTT

7.16 Shared: Burst Splitter → Packet Formation Unit (Read)

Signal	W	Dir	From	To	Handshake	Description
<code>pkt_valid</code>	1	out	Burst Splitter	Read PFU	VALID/READY	Normalized beat context valid
<code>pkt_ready</code>	1	in	Read PFU	Burst Splitter	VALID/READY	Downstream ready
<code>addr[31:0]</code>	32	out	Burst Splitter	Read PFU	VALID/READY	Per-beat read address
<code>TXN_TAG[7:0]</code>	8	out	Burst Splitter	Read PFU	VALID/READY	Transaction tag
<code>pkt_type[1:0]</code>	2	out	Burst Splitter	Read PFU	VALID/READY	01=Read, 11=Err_read
<code>last_beat</code>	1	out	Burst Splitter	Read PFU	VALID/READY	Final beat of sub-transaction
<code>s1_busy</code>	1	out	Burst Splitter	AR Meta FIFO	Level	Stalls Stage 1 FIFO pop during EMIT B state

7.17 Read Path: Packet Formation Unit → Read Packet FIFO

Signal	W	Dir	From	To	Handshake	Description
PKT_DATA[41:0]	42	out	Read PFU	Read Pkt FIFO	FIFO push	Serialized 42-bit read packet: {addr, txn_tag, pkt_type}
PKT_VALID	1	out	Read PFU	Read Pkt FIFO	FIFO push	Packet push strobe

7.18 Read Path: Read Packet FIFO → MC Core

Signal	W	Dir	From	To	Handshake	Description
PKT_DATA[41:0]	42	out	Read Pkt FIFO	MC Core	VALID/READY	Serialized read request packet
PKT_VALID	1	out	Read Pkt FIFO	MC Core	VALID/READY	Packet valid
PKT_READY	1	in	MC Core	Read Pkt FIFO	VALID/READY	MC Core ready; deasserted by ROB watermark

7.19 MC Core Completion Interface (MC Core → CIF)

Signal	W	Dir	From	To	Handshake	Description
RESP_VALID	1	in	MC Core	ROB / Resp Tracker / RTT	VALID/READY (tied)	Completion valid
RESP_READY	1	out	CIF	MC Core	VALID/READY (tied)	Tied permanently high; CIF never stalls MC Core completions
resp_type[1:0]	2	in	MC Core	ROB / Resp Tracker / RTT	VALID/READY (tied)	00=RD_DATA (→ROB, RTT), 01=WR_ACK (→Resp Tracker), 10=ERR

<code>tag[7:0]</code>	8	in	MC Core	ROB / Resp Tracker / RTT	VALID/READY (tied)	Completion tag; [7:2]=AXID, [1:0]=seqnum
<code>data[511:0]</code>	512	in	MC Core	ROB	VALID/READY (tied)	Read data; valid only when <code>resp_type==00</code>
<code>status[3:0]</code>	4	in	MC Core	ROB / Resp Tracker	VALID/READY (tied)	Status/error code; encoding TBD with MC team

7.20 Read Path: ROB → Merge Logic

Signal	W	Dir	From	To	Handshake	Description
<code>data[511:0]</code>	512	out	ROB	Merge Logic	VALID/READY	In-order read data beat emitted by Emit Logic
<code>status[3:0]</code>	4	out	ROB	Merge Logic	VALID/READY	Status for emitted beat
<code>AXID[5:0]</code>	6	out	ROB	Merge Logic	VALID/READY	AXID of emitted beat
<code>rob_last</code>	1	out	RTT (via ROB)	Merge Logic	VALID/READY	Final beat of transaction; sourced from RTT, passed through ROB coincident with final data beat

7.21 Read Path: RTT → ROB / Merge Logic

Signal	W	Dir	From	To	Handshake	Description
<code>rob_last</code>	1	out	RTT	ROB / Merge Logic	Pulse	One-cycle pulse on final MC Core completion for a transaction; <code>cmp_pkts == exp_pkts</code>
<code>RD_FREE_VALID</code>	1	out	RTT	Tag Allocator	Pulse	Read slot free strobe; coincident with <code>rob_last</code>
<code>RD_FREE_TAG[7:0]</code>	8	out	RTT	Tag Allocator	Pulse	Tag being freed; [7:2]=AXID for occupancy decrement

7.22 Read Path: Merge Logic → R Response Generator

Signal	W	Dir	From	To	Handshake	Description
RDATA[511:0]	512	out	Merge Logic	R Resp Gen	VALID/READY	Ordered read data beat
RID[5:0]	6	out	Merge Logic	R Resp Gen	VALID/READY	AXI ID recovered from <code>txn_tag[7:2]</code>
status[3:0]	4	out	Merge Logic	R Resp Gen	VALID/READY	Beat status; drives DECERR when error
RLAST	1	out	Merge Logic	R Resp Gen	VALID/READY	Last beat flag derived from <code>rob_last</code>

7.23 ROB Backpressure Control

Signal	W	Dir	From	To	Handshake	Description
watermark_hit	1	out	ROB Occ Counter	PKT_READY ctrl	Level	Asserted when occupancy ≥ 38 ; combinational from counter
PKT_READY	1	out	ROB	Read Pkt FIFO	Level	Deasserted when <code>watermark_hit</code> high; halts new read requests to MC Core

8 Arbitration Policy

8.1 Overview

Two blocks in CIF are shared across all $N_CLIENTS = 32$ AXI clients: the **Transaction Tag Allocator** and the **Burst Splitter**. Both sit on the common request path after the per-path Request Validators. This section defines how access to these shared resources is arbitrated.

No other block requires explicit arbitration. The AR and AW Metadata FIFOs are partitioned statically (4 slots per client); the ROB, RTT, and Response Tracker are indexed directly by `txn_tag` and require no arbitration. The completion path from MC Core is a single bus fanned out to the ROB, RTT, and Response Tracker simultaneously, with each block filtering on `resp_type` — no arbiter required.

8.2 Transaction Tag Allocator

The Tag Allocator is a single shared instance serving both read and write paths.

Arbitration Mechanism

The Tag Allocator accepts exactly one allocation request per cycle. Upstream priority is resolved as follows:

1. **Write path** and **read path** requests arrive at the Tag Allocator from their respective Request Validators (or Error Handlers). In CIF v1, write and read paths share the Burst Splitter sequentially (single pipeline); the upstream MUX feeding the Tag Allocator selects between write-path and read-path requests using **fixed priority, write > read**. If both paths present a request in the same cycle, the write-path request is accepted; the read-path request is held for the next cycle.
2. Within a single path, only one AXID is presented per cycle (the FIFO head entry). No further arbitration is required at this level.

Per-AXID Stall

Allocation for a given AXID is blocked when `occupancy[AXID] == MAX_OUTSTANDING (4)`. The Stall Logic asserts **STALL** combinationally to the upstream block for that specific AXID only. Other AXIDs are unaffected. **STALL** is released in the cycle after the occupancy counter decrements (i.e., cycle $N + 1$ after `WR_FREE_VALID` or `RD_FREE_VALID` fires for a tag belonging to that AXID).

Error Path

The Error Handler presents allocation requests using the same interface as the Request Validator (`REQ_VALID`, `AXID`). Error-path allocations are subject to the same per-AXID stall and write/read priority rules. No bypass or priority elevation is granted to error transactions.

Simultaneous Free and Allocate

If `alloc_en` and `dec_occ` fire in the same cycle for the same AXID (one slot freed while a new one is being allocated), the occupancy counter holds constant (net-zero change). This is a valid steady-state operating condition and requires no special handling.

8.3 Burst Splitter

The Burst Splitter is a single shared two-stage pipeline instance serving both read and write paths across all clients.

Arbitration Mechanism

Access to the Burst Splitter is implicitly arbitrated by the upstream MUX that feeds the Tag Allocator. Since the Tag Allocator and Burst Splitter are driven from the same upstream MUX with the same priority policy (write > read, one request per cycle), the Burst Splitter processes exactly one transaction at a time.

Pipeline Stall Policy

The Burst Splitter Emit FSM stalls Stage 1 (via `s1_busy`) during EMIT B to prevent the next FIFO entry from being consumed while the second sub-transaction of a split is being emitted. `s1_busy` is deasserted on the cycle the FSM returns to IDLE. Backpressure from downstream (`pkt_ready = 0`) holds the FSM in its current state; `s1_busy` remains asserted until IDLE is reached.

Tag Allocator Stall Folding

If the Tag Allocator asserts STALL before Stage 1 can proceed, the stall is folded into `s1_busy`. Stage 1 does not advance until STALL deasserts and a valid tag is available.

Throughput

Under no-split, no-stall conditions the Burst Splitter achieves one transaction per cycle. A row-boundary split costs one additional cycle per transaction (EMIT A + EMIT B). WRAP bursts iterate over the pre-computed `beat_addrs[]` table; throughput is one beat per cycle per WRAP address, plus one additional cycle if a row-boundary split is also required.

9 Reset and Initialization Sequence

9.1 Overview

CIF supports two reset modes:

- **Power-on reset (POR):** Asserted by the FPGA fabric on configuration. All flip-flops and BRAM init values take effect. CIF remains in reset until the AXI interconnect and MC Core are both ready.
- **Soft reset:** Driven by a register write from a management plane. Must drain the pipeline before asserting; returns CIF to the same state as POR without reconfiguring the FPGA.

All resets are **synchronous, active-high**, qualified on **posedge clk**. There is no asynchronous reset path in CIF.

9.2 Reset Signal

Signal	W	Source	Description
<code>rst_n</code>	1	FPGA top / mgmt plane	Active-low synchronous reset; asserted low to reset all CIF blocks

9.3 Power-On Reset Sequence

The following sequence describes the required initialization order. Steps within the same numbered stage may proceed in parallel.

1. **Assert `rst_n` = 0.** All registered state is cleared synchronously on the next **posedge clk**:
 - AR/AW Metadata FIFO and W Data FIFO: empty, `ARREADY` / `AWREADY` / `WREADY` = 0.
 - Write and Read Packet FIFOs: empty.
 - Tag Allocator: all 64 occupancy counters $\leftarrow$ 0; all 64 seqnum counters $\leftarrow$ 0.
 - Response Tracker register file: all entries cleared (`cmp_pkts` = 0, `err_sticky` = 0, `exp_pkts` = 0).
 - Response Tracker head pointer array: all 64 entries $\leftarrow$ 0.
 - RTT register file: all 256 entries cleared.
 - ROB BRAM: all 64 slots `valid` = 0.
 - ROB free list: all 64 bits $\leftarrow$ 1 (all slots free).
 - ROB lookup table: all 256 entries $\leftarrow$ 0.
 - ROB head pointer array: all 64 registers $\leftarrow$ 0.
 - ROB occupancy counter $\leftarrow$ 0.
 - Burst Splitter Emit FSM $\leftarrow$ IDLE; all internal state registers cleared.
 - Request Validators: combinational; no state to clear.

- Error Handlers: no persistent state to clear.
 - RESP_READY driven high immediately (tied; no reset dependency).
2. **Wait for MC Core ready.** CIF must not issue any packets until MC Core signals it is ready to accept traffic. PKT_VALID must be held low (Packet FIFOs are empty during reset; no additional gating required if FIFOs reset correctly).
 3. **Deassert rst_n = 1.** On the first **posedge** clk after deassertion:
 - AWREADY and ARREADY are driven based on per-client FIFO slot availability. Both are high immediately after reset since all FIFO slots are free.
 - WREADY is driven high immediately (W Data FIFO empty).
 - The Tag Allocator begins accepting allocation requests.
 - The Burst Splitter Emit FSM begins processing from IDLE.
 4. **AXI traffic may begin.** First valid AXI transaction is accepted on the cycle after AWREADY / ARREADY are high and the AXI master presents a valid request.

9.4 Soft Reset Sequence

Soft reset must be preceded by a pipeline drain to avoid losing in-flight transactions.

1. **Quiesce upstream.** The management plane instructs all AXI masters to stop issuing new transactions. New AWVALID and ARVALID are withheld.
2. **Drain the request pipeline.** Wait until:
 - AR and AW Metadata FIFOs are empty.
 - W Data FIFO is empty.
 - Write and Read Packet FIFOs are empty.
 - Burst Splitter Emit FSM is in IDLE.
 - Tag Allocator: all 64 occupancy counters = 0 (no in-flight transactions).
3. **Drain the completion pipeline.** Wait until:
 - ROB occupancy counter = 0 (all read completions have been emitted to Merge Logic).
 - Response Tracker: all 256 register file entries clear (`cmp_pkts == exp_pkts` and freed, or all `exp_pkts == 0`).
 - RTT: all 256 entries clear.
 - No pending BRESP or RRESP outstanding on the AXI B or R channels.
4. **Assert rst_n = 0.** Proceed identically to steps 1–4 of the power-on reset sequence.

9.5 BRAM Initialization Note

The ROB BRAM `valid` bits are the only BRAM state that must be explicitly initialized. On the cycle `rst_n` deasserts, all 256 `valid` bits must read as 0. This is achieved by:

- Using a synchronous-reset flip-flop for each `valid` bit (implemented as distributed RAM or shadow registers alongside the BRAM data), **or**
- Performing a 256-cycle initialization write loop on the BRAM write port during reset, setting `valid = 0` at each address before `rst_n` deasserts.

The second option requires holding `rst_n` low for at least 256 cycles after POR. This is the recommended approach for UltraScale BRAM inference.

10 Error Propagation Timing

10.1 Overview

Address errors detected by the Request Validator are propagated through the normal CIF pipeline as NOP packets using `pkt_type[1:0] = 10 (Err_write)` or `11 (Err_read)`. This section traces the cycle-by-cycle timing of `pkt_type` from error detection to AXI error response, and identifies where `pkt_type` is set, carried, and consumed. Timing counts assume no stalls unless noted.

10.2 Write Path Error Propagation

Cycle	Stage	Event
N	Request Validator	Address Range Check evaluates <code>addr_ok = 0</code> combinationally. <code>ERR_VALID</code> asserted to Error Handler; <code>AXID</code> forwarded. AW Metadata FIFO entry is consumed (popped) in this cycle.
N	Error Handler	Error Handler sets <code>pkt_type = Err_write</code> and presents <code>REQ_VALID + AXID</code> to Tag Allocator in the same cycle.
$N + 1$	Tag Allocator	<code>alloc_en</code> fires (assuming no <code>STALL</code>). <code>TXN_TAG</code> and <code>TAG_VALID</code> registered; valid on cycle $N + 1$. Occupancy counter and seqnum counter update take effect on cycle $N + 1$.
$N + 1$	Burst Splitter Stage 1	<code>pkt_type = Err_write</code> enters Stage 1 alongside <code>TXN_TAG</code> , <code>ADDR</code> , <code>AXLEN</code> , <code>AXSIZE</code> . Stage 1 computes split decision combinationally; pipeline register clocked on cycle $N + 1$.
$N + 2$	Burst Splitter Stage 2	Emit FSM enters EMIT A. <code>pkt_type</code> carried unchanged through Sub-txn A mux. Both sub-transactions (if split) carry the same <code>pkt_type = Err_write</code> .
$N + 2$	Packet Formation Unit	Write PFU assembles 618-bit packet with <code>pkt_type[1:0] = 10</code> in bits [1:0]. <code>data[511:0]</code> and <code>wrapper[63:0]</code> are taken from the W Data FIFO as-is; no drain or discard.
$N + 2$ $N + 2 + F_W$	Write Packet FIFO MC Core	Packet pushed into FIFO. <code>Err_write</code> packet dequeued from Write Packet FIFO and issued to MC Core (<code>PKT_VALID</code> asserted). F_W = Write Packet FIFO occupancy delay (0 under no backpressure). MC Core discards write data; returns error completion (<code>resp_type = 10</code>) on the completion bus.
$N + 2 + F_W + L_{MC}$	Response Tracker	MC Core asserts <code>RESP_VALID</code> with <code>resp_type = 10</code> . Completion Update Logic increments <code>cmp_pkts</code> and sets <code>err_sticky = 1</code> . Compare Logic evaluates <code>cmp_pkts == exp_pkts</code> combinationally (true on this cycle for a single-packet error transaction). <code>txn_done</code> pulses.
$N + 3 + F_W + L_{MC}$	B Response Generator	Emit Logic (registered) drives <code>BRESP_VALID</code> , <code>BID</code> , and <code>BRESP = DECERR</code> to B Response Generator on the cycle after <code>txn_done</code> . <code>BVALID</code> is asserted on the AXI B channel.

L_{MC} denotes MC Core's internal latency from packet receipt to error completion assertion. This is a MC Core parameter and is not constrained by CIF.

Key invariant: `pkt_type` is set once (at the Error Handler output, cycle N) and carried unchanged through the S1→S2 pipeline register and Sub-txn muxes. Neither the Burst Splitter nor the PFU inspect or modify `pkt_type`. Both sub-transactions of a split carry the same `pkt_type`.

10.3 Read Path Error Propagation

Cycle	Stage	Event
N	Request Validator	<code>addr_ok</code> = 0 combinationally. <code>ERR_VALID</code> asserted to Error Handler; <code>AXID</code> forwarded. AR Metadata FIFO entry consumed.
N	Error Handler	<code>pkt_type</code> = <code>Err_read</code> set. <code>REQ_VALID</code> + <code>AXID</code> presented to Tag Allocator.
$N + 1$	Tag Allocator	<code>TXN_TAG</code> and <code>TAG_VALID</code> registered and valid. ROB slot allocated via <code>alloc_req</code> . RTT entry allocated via <code>TAG_VALID</code> path with <code>exp_pkts</code> = <code>ARLEN</code> +1.
$N + 1$	Burst Splitter Stage 1	<code>pkt_type</code> = <code>Err_read</code> enters Stage 1; pipeline register clocked.
$N + 2$	Burst Splitter Stage 2	Emit FSM emits beat(s) with <code>pkt_type</code> = <code>Err_read</code> unchanged.
$N + 2$	Packet Formation Unit	Read PFU assembles 42-bit packet with <code>pkt_type</code> [1:0] = 11 in bits [1:0]. No data field in read packets.
$N + 2$	Read Packet FIFO	Packet pushed into FIFO.
$N + 2 + F_R$	MC Core	<code>Err_read</code> packet issued to MC Core. MC Core returns error completion (<code>resp_type</code> = 10) with no data.
$N + 2 + F_R + L_{MC}$	ROB	Completion Input writes <code>status</code> and <code>valid</code> = 1 to the allocated ROB slot with error status preserved. RTT: <code>cmp_pkts</code> incremented; <code>rob_last</code> pulses (single-beat transaction).
$N + 2 + F_R + L_{MC}$	Merge Logic	Emit Logic emits slot to Merge Logic with <code>status</code> indicating error. <code>rob_last</code> forwarded simultaneously.
$N + 3 + F_R + L_{MC}$	R Response Generator	<code>RVALID</code> asserted with <code>RRESP</code> = <code>DECERR</code> , <code>RID</code> recovered from <code>tag</code> [7:2], <code>RLAST</code> = 1.

10.4 Split Transaction Error Propagation

When an error transaction also requires a row-boundary split (`split_needed` = 1), both sub-transactions carry the same `pkt_type`. The implications are:

- **Write path:** Both `Err_write` packets are issued to MC Core. MC Core returns one error completion per packet. The Response Tracker increments `cmp_pkts` on each completion; `err_sticky` is set on the first. `txn_done` fires only after both completions arrive (`cmp_pkts` == `exp_pkts`). `BRESP` = `DECERR` is emitted once, covering the entire transaction.
- **Read path:** Both `Err_read` packets are issued to MC Core. MC Core returns one error completion per packet (no data). The RTT increments `cmp_pkts` on each; `rob_last` fires after the second. The ROB emits two slots (both with error status); Merge Logic forwards both

to the R Response Generator. **RLAST** is asserted on the final beat; both beats carry **RRESP = DECERR**.

- **Cycle cost of split:** **EMIT B** adds one cycle to the Burst Splitter stage (Stage 2 holds **s1_busy** for one additional cycle). All downstream stages (PFU, FIFO, MC Core, response path) are otherwise unaffected by the split.

10.5 Err.write W Beat Handling

W beats for an **Err.write** transaction flow through the W Data FIFO and Write PFU without modification or draining. The **pkt_type** field in the assembled packet signals to MC Core that the data is invalid; MC Core discards the payload. No special W-beat gating or bypass path exists in CIF. The number of W beats consumed by the PFU is determined by **AWLEN**, consistent with the normal write path.

10.6 Timing Summary

Stage	Write Error Latency	Read Error Latency
Req Validator → Error Handler	0 (combinational)	0 (combinational)
Error Handler → Tag Allocator output	1 cycle	1 cycle
Tag Allocator → Burst Splitter S2 output	1 cycle	1 cycle
Burst Splitter S2 → Packet FIFO push	0 (same cycle)	0 (same cycle)
Packet FIFO → MC Core (no backpressure)	0	0
MC Core latency	L_{MC}	L_{MC}
Final completion → BRESP/RRESP (no split)	1 cycle (Emit Logic reg.)	0 (ROB emit + Merge Logic)
Total CIF-internal latency (no split, no stall)	3 + L_{MC} cycles	2 + L_{MC} cycles

A Burst Splitter

A.1 Overview

The Burst Splitter is a two-stage pipelined block shared across all **N_CLIENTS** AXI clients. It sits between the AW/AR FIFO and the Packet Formation Unit. Its function is to convert validated AXI burst requests into normalized per-beat packets, splitting bursts that cross DRAM row boundaries into two sub-transactions.

Input: AW/AR FIFO output — **ADDR**, **AXID**, **AXLEN**, **AXSIZE**, **AXBURST**

Output: Per-beat context (**addr**, **tag**, **pkt_type**, **beat_cnt**) → Packet Formation Unit via **pkt_valid**

A.2 Stage 1 — Decode & Compute (1 cycle, registered output)

Stage 1 runs combinatorially and captures all results into the S1→S2 pipeline register on the next clock edge. Three sub-blocks operate in parallel:

A.3 Burst Type Decoder

- Input: AXBURST[1:0] from FIFO
- Output: is_wrap / is_incr flag → pipeline register
- FIXED burst type is not supported (design-time assertion)

A.4 Row Boundary Checker

- Inputs: ADDR, AXLEN, AXSIZE from FIFO
- Computes: `addr_start + burst_bytes` vs `ROW_MASK`
- Outputs: `split_needed` flag, `row_boundary` address → Split Point Calc and Beat Count Calc

A.5 WRAP Address Generator

- Inputs: ADDR, AXLEN, AXSIZE from FIFO
- Active only when `is_wrap = 1`
- Outputs: `wrap_boundary`, `beat_addrs[0..N]` → pipeline register
- Maximum 16 entries (AXI4 WRAP burst max `AXLEN = 15`)

A.6 Split Point Calc

- Inputs: `row_boundary` from Row Boundary Checker; `ADDR` from FIFO
- Computes: `addr_b = row_boundary`, `beat_cnt_b = total_beats - beat_cnt_a`
- Output: `addr_b`, `beat_cnt_b` $\rightarrow$ pipeline register

A.7 Beat Count Calc

- Inputs: `row_boundary` from Row Boundary Checker; `AXLEN` from FIFO
- Computes: `beat_cnt_a = beats_to_row_end`, `total_beats = AXLEN + 1`
- Output: `beat_cnt_a`, `total_beats` $\rightarrow$ pipeline register

A.8 S1→S2 Pipeline Register (clocked)

```
{ addr_a, beat_cnt_a, addr_b, beat_cnt_b, split_needed, is_wrap, beat_addrs[], axid,
    pkt_type, tag }
```

A.9 Stage 2 — Emit & Track

Stage 2 is multi-cycle when a split is needed. It stalls Stage 1 (via `s1_busy`) while emitting sub-transaction B.

A.10 Emit FSM

- IDLE $\rightarrow$ EMIT_A
- EMIT_A $\rightarrow$ IDLE (split_needed = 0)
- EMIT_A $\rightarrow$ EMIT_B (split_needed = 1)
- EMIT_B $\rightarrow$ IDLE
- Hold current state if pkt_ready = 0
- Asserts s1_busy during EMIT_B to stall Stage 1

A.11 Sub-transaction A Mux

- Selects: `addr_a`, `beat_cnt_a`, `tag`, `pkt_type` from pipeline register
- Forwards to Pkt Assembler on cycle N

A.12 Sub-transaction B Mux

- Selects: `addr_b`, `beat_cnt_b`, same `tag`, `pkt_type` from pipeline register
- Forwards to Pkt Assembler on cycle N+1

A.13 Beat Tracker

- Maintains `beat_cnt_reg`; decrements on `pkt_accept`
- Asserts `last_beat` flag when count reaches zero
- Output: `last_beat` → Pkt Assembler

A.14 WRAP Beat Address Table

- Up to 16-entry register file, written once on S1→S2 pipeline register transfer
- Input: `beat_addrs[]` from S1→S2 pipeline register
- Read sequentially by `beat_idx` counter
- Output: current beat address → Pkt Assembler

A.15 Pkt Assembler

- Inputs: `addr`, `tag`, `pkt_type`, `beat_cnt` from Sub-txn A/B mux; `last_beat` from Beat Tracker; beat address from WRAP beat addr table
- Assembles normalized read (42-bit) or write (618-bit) packet
- Asserts `pkt_valid` → Packet Formation Unit

A.16 Backpressure & Control Signals

Signal	Direction	Description
<code>s1.busy</code>	Emit FSM → AW/AR FIFO	Stalls Stage 1 from consuming next FIFO entry during <code>EMIT_B</code>
<code>pkt_ready</code>	Packet Formation Unit → Emit FSM	Back-pressure from downstream; FSM holds state when low
<code>pkt_valid</code>	Pkt Assembler → Packet Formation Unit	Indicates normalized packet ready for consumption

A.17 Three Paths Through the Splitter

Path A — No split (INCR, no row crossing)

`split_needed` = 0. Stage 2 emits one packet (`EMIT_A` → `IDLE` in 1 cycle). `s1.busy` never asserted. Throughput: 1 transaction / cycle.

Path B — Row-boundary split (INCR, crosses boundary)

`split_needed = 1`. Stage 1 computes `addr_b`, `beat_cnt_b`. Stage 2: EMIT_A cycle N, asserts `s1_busy`, EMIT_B cycle N+1. Throughput: 1 transaction / 2 cycles.

Path C — WRAP burst

WRAP address generator fills `beat_addrs[]`. Stage 2 iterates via WRAP beat addr table indexed by `beat_idx`. May also split at row boundary.

A.18 Internal State

Field	Description
<code>base_addr</code>	Base address of current burst
<code>beat_idx</code>	Current beat index within burst
<code>beats_left</code>	Remaining beats to emit
<code>burst_type</code>	INCR or WRAP
<code>burst_len</code>	Total beats (AXLEN + 1)
<code>txn_tag</code>	Allocated transaction tag [<code>AXID[5:0]</code> + <code>seqnum[1:0]</code>]

A.19 Timing Notes

- Stage 1 combinatorial depth must close timing in 1 cycle. WRAP address generator is the critical path — limited to $AXLEN \leq 15$ for WRAP bursts (AXI4 constraint).
- S1→S2 register clocked at `posedge clk`.
- `pkt_ready = 0` holds the Emit FSM in its current state; `s1_busy` remains asserted until IDLE is reached.
- Tag Allocator must have a valid tag before Stage 1 can proceed — Tag Allocator stall folds into `s1_busy`.
- Error path (`pkt_type = Err_write / Err_read`): set in Stage 1, carried through pipeline register unchanged. Both sub-transactions on a split carry the same `pkt_type`.

B Request Validator

B.1 Overview

The Request Validator is a purely combinational block instantiated once each on the read and write paths. It pops entries from the AR or AW Metadata FIFO, performs a single runtime check (address range validation), and steers the transaction to one of two downstream paths:

- **Valid path:** Transaction forwarded to the Transaction Tag Allocator with `REQ_VALID` asserted.
- **Error path:** Transaction forwarded to the Error Handler with `ERR_VALID` asserted and `AXID` preserved.

The Request Validator contains no storage, no counters, and no state. All protocol-level checks (burst type, WRAP legality, alignment, burst length) are enforced as design-time assertions in the verification environment only. Compliant AXI4 masters are assumed at runtime.

Parameter	Value	Description
ADDR_W	32	Address width in bits
AXID_W	6	AXI ID width
LEN_W	8	Burst length field width (ARLEN/AWLEN)

Table 33: Request Validator parameters

B.2 Design Parameters

B.3 Internal Blocks

B.4 Address Range Check

- **Type:** Purely combinational.
- **Input:** ADDR[31:0] from the AR/AW Metadata FIFO.
- **Function:** Checks whether ADDR falls within the valid memory address range defined at synthesis time.
- **Output:** addr_ok (1-bit) — asserted when the address is valid, deasserted on an address error.
- **Scope:** Address range check only. No burst type, no alignment, no length checks at runtime.

B.5 Output Mux

- **Type:** Purely combinational.
- **Function:** Routes the transaction to the correct downstream block based on addr_ok.
- **Valid path** (addr_ok == 1): asserts REQ_VALID to the Tag Allocator; forwards AXID[5:0] and ARLEN/AWLEN[7:0].
- **Error path** (addr_ok == 0): asserts ERR_VALID to the Error Handler; forwards AXID[5:0].
- Only one output path is active per transaction. Both paths are never asserted simultaneously.

B.6 Interface Signals

B.7 Inputs

Signal	Width	Source	Description
ADDR[31:0]	32	AR/AW Metadata FIFO	Transaction address
AXID[5:0]	6	AR/AW Metadata FIFO	AXI ID
ARLEN/AWLEN[7:0]	8	AR/AW Metadata FIFO	Burst length
STALL	1	Tag Allocator	Per-AXID stall; holds output until deasserted

Table 34: Request Validator input signals

B.8 Outputs

Signal	Width	Destination	Description
REQ_VALID	1	Tag Allocator	Valid transaction request
AXID[5:0]	6	Tag Allocator	Forwarded AXI ID
ARLEN/AWLEN[7:0]	8	Tag Allocator	Forwarded burst length
ERR_VALID	1	Error Handler	Address error notification
AXID[5:0]	6	Error Handler	AXI ID preserved for NOP tag allocation

Table 35: Request Validator output signals

B.9 Functional Behaviour

1. Request Validator pops the head entry from the AR/AW Metadata FIFO: ADDR[31:0], AXID[5:0], ARLEN/AWLEN[7:0].
2. Address Range Check evaluates `addr_ok` combinationaly.
3. If `addr_ok == 1` and `STALL` is deasserted: Output Mux asserts `REQ_VALID` to the Tag Allocator with `AXID` and `ARLEN/AWLEN`.
4. If `addr_ok == 0`: Output Mux asserts `ERR_VALID` to the Error Handler with `AXID`. The Error Handler sets `pkt_type = Err_write / Err_read` and injects the transaction into the normal pipeline via the Tag Allocator.
5. If `STALL` is asserted (Tag Allocator full for this AXID): the Request Validator holds its outputs stable and does not pop the next FIFO entry until `STALL` deasserts.

B.10 Timing Notes

- Fully combinational. Output (`REQ_VALID` or `ERR_VALID`) is valid in the same cycle as the FIFO head entry is presented.
- `STALL` from the Tag Allocator is also combinational — the combined path (FIFO output → Address Range Check → Output Mux → Tag Allocator Stall Logic → `STALL` back) is a combinational loop that must be timed carefully. A register stage may be inserted between the Request Validator output and the Tag Allocator input if timing closure requires it, at the cost of one cycle of allocation latency.

B.11 Edge Cases

- **Error path stall:** An address error transaction still consumes a Tag Allocator slot. If the AXID is at maximum outstanding (`STALL` asserted), the error path is also held. `ERR_VALID` is not asserted until `STALL` deasserts.
- **Protocol violations:** Illegal burst type, unaligned WRAP, and maximum burst length violations are not checked at runtime. They are enforced as design-time assertions in the verification environment. A non-compliant master will cause undefined behaviour.
- **Simultaneous valid and error:** Not possible — `addr_ok` is a single-bit result; exactly one output path is active per transaction.

C Transaction Tag Allocator

C.1 Overview

The Transaction Tag Allocator is a shared combinational/registered block instantiated once in the CIF and used by both the read and write paths across all `N_CLIENTS = 32` AXI clients.

Its responsibilities are:

- Allocate a unique `txn_tag[7:0]` for every validated transaction entering the pipeline.
- Enforce the per-AXID outstanding transaction limit of `MAX_OUTSTANDING = 4`.
- Stall upstream (Request Validator / Error Handler) on a per-AXID basis when the outstanding limit is reached.
- Free an occupied slot when a transaction completes — via `BRESP` on the write path, or `rob_last` on the read path.

Tag encoding (agreed with MC Core team):

`txn_tag[7:0]`:
[7:2] = `AXID[5:0]` — 6-bit AXI ID, up to 64 unique IDs
[1:0] = `seqnum[1:0]` — per-AXID slot index, wraps at 4

C.2 Design Parameters

Parameter	Value	Description
<code>AXID_W</code>	6	AXI ID width; 64 unique IDs
<code>MAX_OUTSTANDING</code>	4	Max outstanding txns per AXID
<code>SEQNUM_W</code>	2	Seqnum width ($\log_2$ <code>MAX_OUTSTANDING</code>)
<code>TXN_TAG_W</code>	8	Tag width = <code>AXID_W</code> + <code>SEQNUM_W</code>
<code>N_AXIDS</code>	64	Total AXID space (2^6)
<code>OCC_CTR_W</code>	2	Occupancy counter width per AXID

Table 36: Transaction Tag Allocator parameters

C.3 Internal Blocks

C.4 Occupancy Counter Array

- **Structure:** 64 independent 2-bit saturating counters, one per AXID.
- **Increment:** On every successful allocation (`alloc_en` asserted), the counter for the requesting AXID is incremented.
- **Decrement:** On assertion of `WR_FREE_VALID` or `RD_FREE_VALID`, the counter for the AXID extracted from the incoming tag (`tag[7:2]`) is decremented. Each counter supports a 2-wide decrement port: if both `WR_FREE_VALID` and `RD_FREE_VALID` assert in the same cycle for the *same* AXID (different seqnums), the counter decrements by 2 in a single cycle. The counter is implemented as a 3-bit adder with a saturating floor at zero rather than a simple up/down counter, to support this case without a read-modify-write hazard.

- **Read port:** Continuously exposes `occupancy[AXID]` to Stall Logic.

C.5 Seqnum Counter Array

- **Structure:** 64 independent 2-bit wrap counters, one per AXID. Represents the *next* seqnum to be allocated.
- **Increment:** On every successful allocation (`alloc_en` asserted), the counter for the requesting AXID is incremented modulo 4.
- **Read port:** Continuously exposes `seqnum[1:0]` for the requesting AXID to Allocate Logic.
- **No decrement:** Seqnum counters are allocation-only; they are never decremented. The occupancy counter separately tracks the number of in-flight slots.

C.6 Stall Logic

- **Type:** Purely combinational.
- **Function:** Asserts `STALL` for a given AXID when `occupancy[AXID] == MAX_OUTSTANDING(4)`.
- **Granularity:** Per-AXID. Only the AXID currently presented by the upstream block is evaluated; other AXIDs are unaffected.
- **Output:** `STALL` fed back to Request Validator and Error Handler.

C.7 Allocate Logic

- **Gate condition:** Fires only when `REQ_VALID` is asserted and `STALL` is deasserted (`alloc_en = REQ_VALID & ~STALL`).
- **Tag formation:**

$$\text{TXN_TAG}[7:0] = \{ \text{AXID}[5:0], \text{seqnum}[1:0] \}$$
- **Side-effects on alloc:** `inc_occ` to Occupancy Counter Array; `inc_seq` to Seqnum Counter Array.
- **Output:** `TXN_TAG[7:0]` and `TAG_VALID` forwarded to Burst Splitter.
- **Error path:** Error Handler presents AXID with `REQ_VALID` asserted; allocation proceeds identically. The resulting tag is used to form a NOP packet (`pkt.type = Err_write / Err_read`) in the normal pipeline. No special casing in Allocate Logic.

C.8 Free Logic

- **Two free ports:**
 - Write path: `WR_FREE_VALID`, `WR_FREE_TAG[7:0]` — sourced from Response Tracker on `BRESP` emission.
 - Read path: `RD_FREE_VALID`, `RD_FREE_TAG[7:0]` — sourced from Read Transaction Table on `rob_last` assertion.
- **AXID extraction:** `AXID = tag[7:2]` in both cases.

- **Action:** Asserts `dec_occ` to Occupancy Counter Array for the extracted AXID.
- **Simultaneous frees:** Both ports may assert in the same cycle. Two sub-cases:
 - **Different AXIDs:** handled independently, no conflict.
 - **Same AXID, different seqnums:** possible by construction — an AXID may have both outstanding read and write transactions on different seqnums (e.g. `seqnum=0` is a write completing via `WR_FREE`, `seqnum=1` is a read completing via `RD_FREE`, same cycle). Free Logic asserts a dual decrement to the Occupancy Counter Array; the counter decrements by 2 in that cycle via its 2-wide decrement port.
 - **Same AXID, same seqnum:** not possible by construction — a tag is either a read or a write transaction, never both.

C.9 Interface Signals

C.10 Inputs

Signal	Width	Source	Description
REQ_VALID	1	Req Validator / Error Handler	Allocation request valid
AXID[5:0]	6	Req Validator / Error Handler	Requesting AXI ID
WR_FREE_VALID	1	Response Tracker	Write path free strobe
WR_FREE_TAG[7:0]	8	Response Tracker	Tag being freed (write)
RD_FREE_VALID	1	Read Txn Table	Read path free strobe
RD_FREE_TAG[7:0]	8	Read Txn Table	Tag being freed (read)

Table 37: Tag Allocator input signals

C.11 Outputs

Signal	Width	Destination	Description
TXN_TAG[7:0]	8	Burst Splitter	Allocated transaction tag
TAG_VALID	1	Burst Splitter	Tag output valid
STALL	1	Req Validator / Error Handler	Per-AXID stall signal

Table 38: Tag Allocator output signals

C.12 Functional Behaviour

C.13 Allocation Flow

1. Upstream presents `AXID[5:0]` and asserts `REQ_VALID`.
2. Stall Logic reads `occupancy[AXID]` from the Occupancy Counter Array combinatorially.
3. If `occupancy[AXID] < 4`: `STALL` is deasserted, `alloc_en` is asserted.
4. Allocate Logic reads `seqnum[1:0]` from the Seqnum Counter Array for the requesting AXID.
5. `TXN_TAG = {AXID, seqnum}` is driven; `TAG_VALID` is asserted.

6. Occupancy Counter Array increments for AXID.
7. Seqnum Counter Array increments (mod 4) for AXID.

C.14 Stall Flow

1. If `occupancy[AXID] == 4`: STALL is asserted to upstream.
2. Upstream holds `REQ_VALID` and `AXID` stable.
3. Stall is released as soon as Free Logic decrements occupancy for that AXID (i.e., on the next `WR_FREE_VALID` or `RD_FREE_VALID` for a tag belonging to that AXID).

C.15 Free Flow

1. Response Tracker asserts `WR_FREE_VALID` with `WR_FREE_TAG` when a BRESP is emitted (write path).
2. Read Txn Table asserts `RD_FREE_VALID` with `RD_FREE_TAG` when `rob_last` fires (read path).
3. Free Logic extracts `AXID = tag[7:2]` and asserts `dec_occ` to the Occupancy Counter Array.
4. This may immediately release a stall on that AXID if Stall Logic observes the updated occupancy.

C.16 Timing Notes

- Stall Logic is purely combinational: `STALL` is available in the same cycle that `AXID` is presented.
- Allocate Logic is registered: `TXN_TAG` and `TAG_VALID` are valid on the cycle *following* `alloc_en`.
- Counter updates (inc/dec) are registered and take effect on the following clock edge. A free in cycle N releases the stall in cycle $N+1$.

C.17 Edge Cases

- **Simultaneous alloc and free on same AXID:** If `alloc_en` and `dec_occ` fire in the same cycle for the same AXID, occupancy is held constant (net zero change). This is a valid steady-state operating condition.
- **Simultaneous dual-free on same AXID:** If both `WR_FREE_VALID` and `RD_FREE_VALID` assert in the same cycle for the same AXID, the Occupancy Counter decrements by 2. This requires the counter to be implemented as a sum-based update rather than a simple up/down counter. A design-time assertion should verify that occupancy never goes below zero after a dual decrement — guaranteed by construction since two in-flight tags on the same AXID implies occupancy ≥ 2 at that moment.
- **Error path:** Error Handler allocates a tag identically to valid transactions. The resulting tag is embedded in a NOP packet (`pkt_type = Err.write / Err.read`). The slot is freed normally on MC Core error completion.
- **Underflow protection:** The Occupancy Counter Array must not decrement below zero. This is guaranteed by construction — a free is only issued when a tag is in-flight — but a design-time assertion should be included in the verification environment.

D Response Tracker

D.1 Overview

The Response Tracker is a write-path block responsible for aggregating per-packet MC Core completions into a single AXI B channel response per transaction. It tracks every outstanding write transaction from tag allocation to final BRESP emission.

Its responsibilities are:

- Store per-transaction metadata at allocation time: expected packet count, completed packet count, error status.
- Increment the completed packet counter on each `WR.ACK` completion received from MC Core.
- Assert `txn_done` when all expected packets for a transaction have completed.
- Enforce AXI4 B channel ordering: BRESP is emitted in-order per AXID using a per-AXID head pointer.
- Propagate error status: if any packet in a burst returns an error completion, the sticky error bit is set and the transaction generates `BRESP = DECERR`.
- Free the occupied Tag Allocator slot on BRESP emission via the `WR.FREE` port.

Note on abstraction levels. The full CIF pipeline diagram shows MC Core as the sole input to the Response Tracker. At the internal logic level, the Tag Allocator also provides an allocation signal — this is a resource management path (writing `exp_pkts` at tag assignment time) that is separate from the data completion path. Both are correct; they operate at different abstraction levels.

D.2 Design Parameters

Parameter	Value	Description
AXID_W	6	AXI ID width; 64 unique IDs
SEQNUM_W	2	Seqnum width
TXN_TAG_W	8	Tag width = AXID_W + SEQNUM_W
RT_DEPTH	256	Register file entries = 2^8
EXP_PKTS_W	8	Expected packets width (covers AWLEN+1, max 256)
CMP_PKTS_W	8	Completed packets counter width
N_AXIDS	64	Head pointer array depth
HEAD_W	8	Head pointer width = TXN_TAG_W

Table 39: Response Tracker parameters

D.3 Internal Blocks

D.4 Register File

- **Structure:** 256-entry register file. Each entry stores:
 - `exp_pkts[7:0]` — expected packet count (= AWLEN+1)

- `cmp_pkts[7:0]` — completed packet count
- `status[3:0]` — MC Core status code from most recent error completion
- `err_sticky[0]` — sticky error bit; set on first error completion, never cleared until entry is freed
- **Addressing:** `addr[7:0] = TXN_TAG[7:0] = {AXID[5:0], seqnum[1:0]}`. Tag directly indexes the register file — no translation required.
- **Write ports:**
 - Alloc Write Logic: writes `exp_pkts`, initialises `cmp_pkts = 0`, `err_sticky = 0` on allocation.
 - Completion Update Logic: increments `cmp_pkts`, optionally sets `err_sticky` and `status` on each `WR_ACK` completion.
 - Free Logic: clears the entry on `BRESP` emission.
- **Read ports:** Continuously exposes `cmp_pkts` and `exp_pkts` to Compare Logic; exposes `err_sticky` and `status` to Emit Logic.

D.5 Alloc Write Logic

- Activated on `TAG_VALID` from the Tag Allocator.
- Receives `TXN_TAG[7:0]` and `AWLEN[7:0]`.
- Computes `exp_pkts = AWLEN + 1`.
- Issues `write_en` to the Register File at address `TXN_TAG[7:0]`, writing `exp_pkts` and initialising `cmp_pkts = 0`, `err_sticky = 0`.

D.6 Completion Update Logic

- Taps the MC Core completion bus: `RESP_VALID`, `resp_type[1:0]`, `tag[7:0]`, `status[3:0]`.
- **Filter:** acts only when `resp_type == 2'b01` (`WR_ACK`). `RD_DATA` completions (`resp_type == 2'b00`) are destined for the ROB and RTT and are ignored here.
- On a normal `WR_ACK` (`resp_type == 2'b01`, no error): asserts `inc_en` to the Register File at address `tag[7:0]`, incrementing `cmp_pkts` by 1.
- On an error completion (`resp_type == 2'b10`): increments `cmp_pkts` and additionally sets `err_sticky = 1` and latches `status[3:0]`. The sticky bit is never cleared by subsequent completions — one error poisons the entire transaction.

D.7 Compare Logic

- Purely combinational.
- Reads `cmp_pkts[7:0]` and `exp_pkts[7:0]` from the Register File for the entry currently being updated.
- Asserts `txn_done` as a one-cycle pulse when `cmp_pkts == exp_pkts`.

- `txn_done` is forwarded to Emit Logic to trigger BRESP evaluation.

D.8 Head Pointer Array

- **Structure:** 64×8 -bit registers, one per AXID. Each register holds the `TXN_TAG` of the next transaction expected to emit BRESP for that AXID.
- **Init:** Written on first allocation for an AXID (occupancy $0 \rightarrow 1$). On subsequent allocations for the same AXID, the head pointer is not updated — it is owned by Emit Logic from that point forward.
- **Advance:** Emit Logic increments the head pointer (mod 4 on the seqnum field) after each BRESP is emitted for that AXID.
- **Purpose:** Enforces AXI4 B channel ordering — BRESP is only emitted for the transaction at the head, not for any later-completing transaction of the same AXID.

D.9 Emit Logic

- Checks whether the transaction at `head_tag[AXID]` is done: reads `txn_done` from Compare Logic and confirms the completing tag matches the current head pointer for that AXID.
- If head slot is done:
 - Reads `err_sticky` and `status` from the Register File.
 - Asserts `BRESP_VALID` to the B Response Generator.
 - Drives `BID[5:0] = head_tag[7:2]` (AXID recovered from tag).
 - Drives `BRESP[1:0] = OKAY` if `err_sticky == 0`; `DECERR` if `err_sticky == 1`.
 - Asserts `advance_head` to the Head Pointer Array for the relevant AXID.
 - Asserts `emit_done` with `tag[7:0]` to Free Logic.
- If head slot is not yet done: Emit Logic holds. Later-completing transactions for the same AXID are buffered in the Register File until the head slot completes.

D.10 Free Logic

- Activated by `emit_done` from Emit Logic.
- Asserts `clear_en` to the Register File at address `tag[7:0]`, clearing the entry in the same cycle as BRESP emission.
- Simultaneously asserts `WR_FREE_VALID` and drives `WR_FREE_TAG[7:0]` to the Tag Allocator's `WR_FREE` port, decrementing the occupancy counter for the freed AXID.

D.11 Interface Signals

D.12 Inputs

Signal	Width	Source	Description
<code>TXN_TAG[7:0]</code>	8	Tag Allocator	Allocated tag at write transaction time

AWLEN[7:0]	8	Tag Allocator	Burst length; <code>exp_pkts</code> = AWLEN+1
TAG_VALID	1	Tag Allocator	Allocation strobe
RESP_VALID	1	MC Core	Completion valid
resp_type[1:0]	2	MC Core	00=RD_DATA (ignored), 01=WR_ACK, 10=ERR
tag[7:0]	8	MC Core	Completion tag; used as register file address
status[3:0]	4	MC Core	Error status; latched on error completion

Table 40: Response Tracker input signals

D.13 Outputs

Signal	Width	Destination	Description
BRESP_VALID	1	B Response Generator	BRESP ready to emit
BID[5:0]	6	B Response Generator	AXI ID recovered from <code>head_tag[7:2]</code>
BRESP[1:0]	2	B Response Generator	OKAY or DECERR
WR_FREE_VALID	1	Tag Allocator (WR_FREE port)	Free strobe; coincident with BRESP emit
WR_FREE_TAG[7:0]	8	Tag Allocator (WR_FREE port)	Tag being freed

Table 41: Response Tracker output signals

D.14 Internal Signals

Signal	Width	From	To	Description
<code>write_en</code>	1	Alloc Logic	Write Register File	Write strobe on allocation
<code>wr_addr[7:0]</code>	8	Alloc Logic	Write Register File	= <code>TXN_TAG[7:0]</code>
<code>exp_pkts[7:0]</code>	8	Alloc Logic	Write Register File	= AWLEN+1
<code>inc_en</code>	1	Completion Update Logic	Register File	Increment <code>cmp_pkts</code> strobe
<code>err_set</code>	1	Completion Update Logic	Register File	Set <code>err_sticky</code> on error
<code>inc_addr[7:0]</code>	8	Completion Update Logic	Register File	= <code>tag[7:0]</code> from MC Core
<code>cmp_pkts[7:0]</code>	8	Register File	Compare Logic	Current completed count
<code>exp_pkts[7:0]</code>	8	Register File	Compare Logic	Expected count (read port)
<code>err_sticky</code>	1	Register File	Emit Logic	Sticky error flag
<code>txn_done</code>	1	Compare Logic	Emit Logic	Pulse: all packets completed
<code>head_tag[7:0]</code>	8	Head Pointer Array	Emit Logic	Next tag to emit per AXID

<code>advance_head</code>	1	Emit Logic	Head Pointer Array	Increment head pointer for AXID
<code>emit_done</code>	1	Emit Logic	Free Logic	BRESP emitted; trigger free
<code>clear_en</code>	1	Free Logic	Register File	Clear entry strobe
<code>clear_addr[7:0]</code>	8	Free Logic	Register File	= tag of emitted transaction

Table 42: Response Tracker internal signals

D.15 Functional Behaviour

D.16 Allocation Flow

1. Tag Allocator asserts `TAG_VALID` with `TXN_TAG[7:0]` and `AWLEN[7:0]`.
2. Alloc Write Logic computes `exp_pkts = AWLEN + 1`.
3. Register File is written at address `TXN_TAG[7:0]`: `exp_pkts` stored, `cmp_pkts = 0`, `err_sticky = 0`.
4. If this is the first in-flight transaction for this AXID (occupancy 0→1), Head Pointer Array is initialised with `TXN_TAG[7:0]`.

D.17 Completion Flow

1. MC Core asserts `RESP_VALID` with `resp_type == 2'b01` (`WR_ACK`) and `tag[7:0]`.
2. Completion Update Logic asserts `inc_en` at address `tag[7:0]`: `cmp_pkts` incremented by 1.
3. If `resp_type == 2'b10` (`ERR`): additionally sets `err_sticky = 1` and latches `status[3:0]`.
4. Compare Logic evaluates `cmp_pkts == exp_pkts` combinationally; asserts `txn_done` if true.

D.18 Emit Flow

1. `txn_done` pulses for a completing transaction.
2. Emit Logic checks whether the completing tag matches `head_tag` for that AXID.
3. If match (in-order): reads `err_sticky` from Register File, drives `BRESP_VALID`, `BID`, and `BRESP` (`OKAY` or `DECERR`) to B Response Generator. Asserts `advance_head` and `emit_done`.
4. If no match (out-of-order): transaction is complete in the Register File but the head has not yet completed. Emit Logic holds. `BRESP` is deferred until the head slot completes.

D.19 Free Flow

1. `emit_done` triggers Free Logic in the same cycle as `BRESP` emission.
2. Free Logic asserts `clear_en` to Register File: entry is invalidated.
3. `WR_FREE_VALID` and `WR_FREE_TAG` are asserted to Tag Allocator, decrementing occupancy for the freed AXID.

D.20 Timing Notes

- Alloc Write Logic is registered: Register File write takes effect the cycle after `TAG_VALID`.
- Completion Update Logic is registered: `cmp_pkts` increment takes effect the cycle after `RESP_VALID`.
- Compare Logic is combinational: `txn_done` appears in the same cycle the increment is visible — one cycle after the final `RESP_VALID`.
- Emit Logic is registered: `BRESP_VALID` is asserted the cycle after `txn_done` is evaluated.
- Free Logic fires in the same cycle as BRESP emission: `clear_en` and `WR_FREE_VALID` are asserted without additional latency.
- Simultaneous allocation and completion on the same tag are not possible by construction — a tag is only re-allocated after its slot is freed, which occurs after BRESP emission.

D.21 Edge Cases

- **Single-packet transaction (`AWLEN = 0`):** `exp_pkts = 1`. `txn_done` fires on the first and only `WR_ACK`. Handled identically to multi-packet transactions.
- **Partial error in a burst:** If one packet in a burst returns `ERR` and others return `WR_ACK`, `err_sticky` is set on the first error. All subsequent completions still increment `cmp_pkts` normally. On BRESP emission, `BRESP = DECERR` regardless of how many packets succeeded.
- **Out-of-order completion within an AXID:** AXI4 requires B channel responses to be in-order per AXID. If a later transaction of an AXID completes before the head transaction, Emit Logic holds the BRESP until the head slot is done. The later transaction's completion is safely held in the Register File.
- **RD_DATA completions:** Ignored entirely by Completion Update Logic. These are processed by the ROB and RTT on the read path.
- **Over-completion protection:** `cmp_pkts` must never exceed `exp_pkts`. Guaranteed by construction since MC Core returns exactly one `WR_ACK` per issued write packet. A design-time assertion should be included in the verification environment.
- **Simultaneous alloc and completion on same AXID (different seqnums):** These target different register file entries (different `tag[7:0]` addresses) — no conflict.

E Read Transaction Table (RTT)

E.1 Overview

The Read Transaction Table (RTT) is a per-transaction tracking structure for the read path. It is allocated at tag assignment time and tracks completion progress for every outstanding read transaction. When all expected packets for a transaction have been received from MC Core, the RTT asserts `rob_last`, signalling the final beat to the ROB and Merge Logic. The entry is freed in the same cycle.

The RTT operates independently of the ROB — both blocks tap the same MC Core completion bus, but serve different purposes: the ROB stores and reorders data; the RTT tracks transaction-level progress and drives `rob_last`.

E.2 Design Parameters

Parameter	Value	Description
AXID_W	6	AXI ID width
SEQNUM_W	2	Seqnum width
TXN_TAG_W	8	Tag width = AXID_W + SEQNUM_W
RTT_DEPTH	256	Entries = $2^{\text{AXID_W} + \text{SEQNUM_W}}$
EXP_PKTS_W	8	Expected packets width (covers ARLEN+1, max 256)
CMP_PKTS_W	8	Completed packets counter width

Table 43: RTT parameters

E.3 Internal Blocks

E.4 RTT Register File

- **Structure:** 256-entry register file. Each entry stores:
 - AXID[5:0] — AXI ID of the transaction
 - `exp_pkts`[7:0] — expected packet count (= ARLEN+1)
 - `cmp_pkts`[7:0] — completed packet count (incremented on each MC Core completion)
- **Addressing:** `addr`[7:0] = `TXN_TAG`[7:0] = {AXID[5:0], `seqnum`[1:0]}. The tag directly indexes the register file — no translation required.
- **Write ports:**
 - Alloc Write Logic writes `exp_pkts` on allocation.
 - Completion Update Logic increments `cmp_pkts` on each RD_DATA completion.
 - Free Logic clears the entry (zeroes `cmp_pkts`, invalidates) on `rob_last`.
- **Read port:** Continuously exposes `cmp_pkts` and `exp_pkts` to Compare Logic.

E.5 Alloc Write Logic

- Activated on TAG_VALID from the Tag Allocator.
- Receives `TXN_TAG`[7:0] and `ARLEN`[7:0].
- Computes `exp_pkts` = ARLEN + 1.
- Issues `write_en` to the RTT Register File at address `TXN_TAG`[7:0], writing `exp_pkts` and initialising `cmp_pkts` = 0.

E.6 Completion Update Logic

- Taps the MC Core completion bus: `RESP_VALID`, `resp_type[1:0]`, `tag[7:0]`.
- **Filter:** acts only when `resp_type == 2'b00` (`RD_DATA`). `WR_ACK` completions (`resp_type == 2'b01`) are destined for the Response Tracker and are ignored. `ERR` completions for read transactions (`resp_type == 2'b10`) are treated as a valid completion event — `cmp_pkts` is still incremented so the transaction reaches `rob_last` and completes normally; error status propagation is handled by the ROB and R Response Generator.
- Asserts `inc_en` to the RTT Register File at address `tag[7:0]`, incrementing `cmp_pkts` by 1.

E.7 Compare Logic

- Purely combinational.
- Reads `cmp_pkts[7:0]` and `exp_pkts[7:0]` from the RTT Register File for the entry currently being updated.
- Asserts `rob_last` as a one-cycle pulse when `cmp_pkts == exp_pkts`.
- `rob_last` is forwarded to two destinations simultaneously: Free Logic (to clear the entry) and the ROB (passed through to Merge Logic coincident with the final data beat).

E.8 Free Logic

- Activated by `rob_last` from Compare Logic.
- Asserts `clear_en` to the RTT Register File at address `tag[7:0]`, clearing the entry in the same cycle as `rob_last`.
- Simultaneously asserts `RD_FREE_VALID` and drives `RD_FREE_TAG[7:0]` to the Tag Allocator's `RD_FREE` port, triggering occupancy decrement for the freed AXID.

E.9 Interface Signals

E.10 Inputs

Signal	Width	Source	Description
<code>TXN_TAG[7:0]</code>	8	Tag Allocator	Allocated tag; indexes the register file
<code>ARLEN[7:0]</code>	8	Tag Allocator	Burst length; <code>exp_pkts = ARLEN+1</code>
<code>TAG_VALID</code>	1	Tag Allocator	Allocation strobe
<code>RESP_VALID</code>	1	MC Core	Completion valid
<code>resp_type[1:0]</code>	2	MC Core	00= <code>RD_DATA</code> , 01= <code>WR_ACK</code> , 10= <code>ERR</code> ; RTT acts on <code>RD_DATA</code> and <code>ERR</code>
<code>tag[7:0]</code>	8	MC Core	Completion tag; used as register file address

Table 44: RTT input signals

E.11 Outputs

Signal	Width	Destination	Description
<code>rob_last</code>	1	ROB / Merge Logic	Pulse on final packet of a transaction
<code>RD_FREE_VALID</code>	1	Tag Allocator (RD_FREE port)	Free strobe; coincident with <code>rob_last</code>
<code>RD_FREE_TAG[7:0]</code>	8	Tag Allocator (RD_FREE port)	Tag being freed

Table 45: RTT output signals

E.12 Internal Signal Table

Signal	Width	From	To	Description
<code>write_en</code>	1	Alloc Logic	Write RTT Reg File	Write strobe on allocation
<code>wr_addr[7:0]</code>	8	Alloc Logic	Write RTT Reg File	= <code>TXN_TAG[7:0]</code>
<code>exp_pkts[7:0]</code>	8	Alloc Logic	Write RTT Reg File	= <code>ARLEN+1</code>
<code>inc_en</code>	1	Completion Update Logic	RTT Reg File	Increment <code>cmp_pkts</code> strobe
<code>inc_addr[7:0]</code>	8	Completion Update Logic	RTT Reg File	= <code>tag[7:0]</code> from MC Core
<code>cmp_pkts[7:0]</code>	8	RTT Reg File	Compare Logic	Current completed count
<code>exp_pkts[7:0]</code>	8	RTT Reg File	Compare Logic	Expected count (read port)
<code>rob_last</code>	1	Compare Logic	Free Logic, ROB	One-cycle pulse on final completion
<code>clear_en</code>	1	Free Logic	RTT Reg File	Clear entry strobe
<code>clear_addr[7:0]</code>	8	Free Logic	RTT Reg File	= <code>tag[7:0]</code> of completing txn

Table 46: RTT internal signals

E.13 Functional Behaviour

E.14 Allocation Flow

1. Tag Allocator asserts `TAG_VALID` with `TXN_TAG[7:0]` and `ARLEN[7:0]`.
2. Alloc Write Logic computes `exp_pkts = ARLEN + 1`.
3. RTT Register File is written at address `TXN_TAG[7:0]`: `exp_pkts` is stored, `cmp_pkts` is initialised to zero.

E.15 Completion Flow

1. MC Core asserts `RESP_VALID` with `resp_type == 2'b00` (`RD_DATA`) and `tag[7:0]`.

2. Completion Update Logic asserts `inc_en` at address `tag[7:0]`: `cmp_pkts` is incremented by 1.
3. Compare Logic reads the updated `cmp_pkts` and `exp_pkts` combinationaly.
4. If `cmp_pkts == exp_pkts`: `rob_last` is asserted as a one-cycle pulse.

E.16 Free Flow

1. `rob_last` pulse triggers Free Logic in the same cycle.
2. Free Logic asserts `clear_en` to the RTT Register File: the entry is invalidated.
3. Free Logic simultaneously asserts `RD_FREE_VALID` with `RD_FREE_TAG` to the Tag Allocator, decrementing the occupancy counter for the freed AXID.

E.17 rob_last Routing

`rob_last` is a sideband signal to Merge Logic. It travels in parallel with, not through, the ROB. The ROB emits `data[511:0]`, `status[3:0]`, and `AXID[5:0]` to Merge Logic; the RTT emits `rob_last` to Merge Logic coincident with the final data beat. Merge Logic combines both to assert `RLAST` on the AXI R channel. There is no feedback path from the RTT into the ROB's internal logic.

E.18 Timing Notes

- Alloc Write Logic is registered: the register file write takes effect on the cycle following `TAG_VALID`.
- Completion Update Logic is registered: `cmp_pkts` increment takes effect on the cycle following `RESP_VALID`.
- Compare Logic is combinational: `rob_last` is available in the same cycle as the `cmp_pkts` update is visible. In practice this means `rob_last` appears one cycle after the final `RESP_VALID`.
- Free Logic fires in the same cycle as `rob_last`: the register file entry is cleared and `RD_FREE_VALID` is asserted without additional latency.
- Simultaneous allocation and completion on the same tag are not possible by construction — a tag is only allocated after its previous occupancy slot is freed, and a completion can only arrive for an in-flight tag.

E.19 Edge Cases

- **Single-beat transaction (`ARLEN = 0`):** `exp_pkts = 1`. `rob_last` is asserted on the first and only completion. Handled identically to multi-beat transactions.
- **Error completion:** `resp_type == 2'b10` (ERR) for a read transaction is treated as a valid completion event. `cmp_pkts` is incremented normally so the transaction progresses to `rob_last`. Error status is stored in the ROB slot and forwarded to the R Response Generator as `DECERR`. The RTT does not need to distinguish error from normal completions.
- **WR_ACK completions:** `resp_type == 2'b01` completions are ignored by the RTT entirely. They are processed by the Response Tracker on the write path.

- **Over-completion protection:** `cmp_pkts` must never exceed `exp_pkts`. This is guaranteed by construction since MC Core returns exactly one completion per issued read packet. A design-time assertion should be included in the verification environment.

F Reorder Buffer (ROB)

F.1 Overview

The Reorder Buffer (ROB) absorbs out-of-order read completions from MC Core and emits them in-order per AXID to Merge Logic. Depth is tied to `MC_RD_BUF` = 100 (MC Core confirmed maximum in-flight pipeline depth). Since `RESP_READY` must always be held high, the ROB can never stall incoming completions — backpressure is instead applied by halting new read requests when occupancy reaches the watermark threshold.

Input Signals

Signal	Width	Source	Description
<code>alloc_req</code>	1	Tag Allocator	Pulse: allocate a new ROB slot
<code>AXID[5:0]</code>	6	Tag Allocator	AXI ID of transaction being allocated
<code>seqnum[1:0]</code>	2	Tag Allocator	Seqnum of transaction being allocated
<code>tag[7:0]</code>	8	Tag Allocator	Full tag (used to write Lookup Table)
<code>RESP_VALID</code>	1	MC Core	Completion valid strobe
<code>resp_type[1:0]</code>	2	MC Core	00=RD_DATA, 01=WR_ACK, 10=ERR; ROB acts only on RD_DATA
<code>tag[7:0]</code>	8	MC Core	Completion tag; [7:2]=AXID, [1:0]=seqnum
<code>data[511:0]</code>	512	MC Core	Read data payload
<code>status[3:0]</code>	4	MC Core	Completion status/error code

Table 47: ROB input signals

Output Signals

Signal	Width	Destination	Description
<code>data[511:0]</code>	512	Merge Logic	In-order read data beat
<code>status[3:0]</code>	4	Merge Logic	Status for emitted beat
<code>AXID[5:0]</code>	6	Merge Logic	AXID of emitted beat
<code>rob_last</code>	1	Merge Logic	Asserted on final beat of a transaction; sourced from RTT, passed through
<code>PKT_READY</code>	1	Packet FIFO	Deasserted when watermark hit; halts new read requests
<code>RESP_READY</code>	1	MC Core	Tied high permanently

Table 48: ROB output signals

F.2 Blocks

F.3 Free List

- 64-bit bitmask. 1 = free, 0 = occupied.
- Allocate: priority encoder selects lowest free bit, clears it, outputs `free_slot_idx[5:0]`.
- Free: Emit Logic sets the bit at `slot_idx` via `free_req`.
- Outputs `free_slot_idx` to Lookup Table (write), ROB BRAM (init), and Head Pointer Array (first allocation).

F.4 Occupancy Counter

- Up/down counter, range 0–64.
- Incremented by `alloc_req` from Free List.
- Decremented by `free_req` from Emit Logic.
- Asserts `watermark_hit` when count $\geq$ `ROB_DEPTH` – `ROB_WATERMARK` = 128.

F.5 Lookup Table

- 256-entry register file, addressed by `{AXID[5:0], seqnum[1:0]}` (8-bit address).
- Each entry stores `slot_idx[5:0]`.
- Write port: Tag Allocator writes `slot_idx` at `{AXID, seqnum}` on allocation.
- Read port: Completion Input reads `slot_idx` using `{tag[7:2], tag[1:0]}`.
- Clear: Emit Logic clears entry at `{AXID, seqnum}` after emit.

F.6 ROB BRAM

- Dual-port BRAM, 64×517 bits per slot: `valid[0]`, `data[511:0]`, `status[3:0]`.
- Write port: Completion Input writes `data`, `status`, `valid=1` at `slot_idx`.
- Write port (init): Free List writes `valid=0` at `free_slot_idx` on allocation.
- Read port: Head Pointer Array supplies `head[AXID]` as read address; all 64 head slots checked in parallel each cycle.
- 1-cycle read latency — Emit Logic must account for this in its valid check timing.

F.7 Completion Input

- Receives `RESP_VALID`, `resp_type[1:0]`, `tag[7:0]`, `data[511:0]`, `status[3:0]` from MC Core.
- **Filters on `resp_type`:** acts only when `resp_type == 2'b00` (RD_DATA). Completions with `resp_type == 2'b01` (WR_ACK) are destined for the Response Tracker and are ignored by the ROB. ERR completions (`resp_type == 2'b10`) for read transactions are written into the ROB with the error status preserved so DECERR can be forwarded to the R Response Generator.
- Extracts `AXID = tag[7:2]`, `seqnum = tag[1:0]`.
- Reads `slot_idx` from Lookup Table using `{AXID, seqnum}`.
- Writes `data`, `status`, `valid=1` to ROB BRAM at `slot_idx`.

F.8 Head Pointer Array

- 64 registers of 6 bits each, one per AXID. Stores slot index of next expected beat.

- Written on first allocation only: `head[AXID] = free_slot_idx` when the AXID has no in-flight slots (occupancy transitions 0→1 for that AXID). On subsequent allocations for the same AXID, `head[AXID]` is *not* updated — it is owned exclusively by Emit Logic from that point forward.
- Advanced by Emit Logic after each emit: Emit Logic computes `next_seqnum = (current_seqnum + 1) % 4`, then reads `slot_idx` for `{AXID, next_seqnum}` from the Lookup Table and writes it into `head[AXID]`. The head pointer stores a slot index, not a seqnum — direct increment is incorrect since the Free List assigns slot indices non-contiguously.
- Reset to `free_slot_idx` from the Free List only when all in-flight slots for an AXID have been emitted and a new allocation arrives (per-AXID occupancy returns to 0 then rises again to 1).
- All 64 head pointers drive the ROB BRAM read port simultaneously each cycle.

F.9 Emit Logic

- Reads `valid` bits from all 64 head slots out of ROB BRAM each cycle.
- Priority encoder selects lowest AXID whose head slot has `valid=1`.
- On selection: reads `data` and `status` from BRAM; emits to Merge Logic along with AXID and `rob_last`. `rob_last` is *not* generated inside the ROB — it is sourced from the Read Transaction Table (RTT), which asserts it when `completed_packets == expected_packets` for that transaction. The ROB passes it through to Merge Logic coincident with the final data beat.
- After emit:
 - Clears `valid` in BRAM slot.
 - Advances `head[AXID]` in Head Pointer Array.
 - Sends `free_req + slot_idx` to Free List (sets bitmask bit).
 - Decrements Occupancy Counter via `free_req`.
 - Clears Lookup Table entry at `{AXID, seqnum}`.
- Maximum one emission per cycle (single R channel path).

F.10 PKT_READY Control

- Purely combinational.
- Asserts `PKT_READY=0` to Packet FIFO when `watermark_hit` is high.
- Halts new read requests to MC Core; prevents ROB overflow since `RESP_READY` cannot be deasserted.

F.11 Allocation Path

1. Tag Allocator sends `alloc_req` (pulse) with AXID, `seqnum`, `tag`.
2. Free List pops `free_slot_idx` (priority encoder on bitmask), clears the bit.
3. Free List writes `slot_idx` to Lookup Table at `{AXID, seqnum}`.
4. Free List initializes ROB BRAM slot: `valid=0` at `free_slot_idx`.
5. Free List stores `free_slot_idx` into `head[AXID]` (first allocation for that AXID only).
6. Occupancy Counter increments.

F.12 Completion Path

1. MC Core asserts `RESP_VALID` with `tag[7:0]`, `data[511:0]`, `status[3:0]`.
2. Completion Input extracts `AXID = tag[7:2]`, `seqnum = tag[1:0]`.

3. Completion Input reads `slot_idx` from Lookup Table.
4. Completion Input writes `data`, `status`, `valid=1` to ROB BRAM at `slot_idx`.
5. `RESP_READY` is tied high — MC Core never stalled on this path.

F.13 Emit Path

1. Each cycle, all 64 `head[AXID]` values drive the ROB BRAM read port.
2. BRAM returns `valid` bits for all 64 head slots (1-cycle latency).
3. Priority encoder in Emit Logic selects lowest AXID with `valid=1`.
4. Emit Logic reads `data` and `status` from selected slot; forwards to Merge Logic with `AXID` and `last_beat`.
5. Slot is freed: `valid` cleared, Free List bit set, Occupancy Counter decremented, Lookup Table entry cleared.
6. `head[AXID]` advanced: Emit Logic reads `slot_idx` for `{AXID, next_seqnum}` from Lookup Table and writes into `head[AXID]`. This read occurs before the Lookup Table clear in the same cycle — clear only affects the just-emitted `current_seqnum` entry, not `next_seqnum`.

F.14 Backpressure & Overflow Prevention

Signal	Description
<code>watermark_hit</code>	Asserted when occupancy ≥ 128 . Drives <code>PKT_READY</code> control.
<code>PKT_READY=0</code>	Stops Packet FIFO from issuing new read requests to MC Core.
<code>RESP_READY=1</code>	Tied high permanently. ROB can never stall MC Core completions.

`ROB_WATERMARK` = 128, derived from MC Core’s confirmed maximum pipeline depth of 100, rounded up conservatively to account for scheduler speculation and forward pressure. This headroom ensures all completions in-flight within MC Core after `PKT_READY` deasserts can return without overflowing the ROB.

F.15 Key Parameters

Parameter	Value
ROB depth	256 entries
Slot width	517 bits (1 valid + 512 data + 4 status)
Free list width	256 bits
Lookup table	256 entries $\times$ 8-bit slot index
Head pointers	64 $\times$ 8-bit registers
Watermark threshold	128 (<code>ROB_DEPTH</code> – <code>ROB_WATERMARK</code> = 256 – 128)
<code>MC_RD_BUF</code>	100 (MC Core confirmed max pipeline depth)
<code>ROB_WATERMARK</code>	128 (conservative; covers speculation and forward pressure)
Max emissions	1 per cycle

F.16 Timing Notes

- ROB BRAM has 1-cycle read latency. Emit Logic valid check must be pipelined accordingly — register the read address, use valid bit one cycle after presenting `head[AXID]`.
- Priority encoder across 64 valid bits is the critical path in Emit Logic. May need pipelining at high frequencies.
- Allocation and completion can occur in the same cycle — write port (completion) and init write (allocation) are independent BRAM ports.
- `watermark_hit` is combinational from the Occupancy Counter — no additional latency before `PKT_READY` deasserts.
- Simultaneous `alloc_req` and `free_req` on the Free List bitmask in the same cycle is safe — they always target different bits (alloc clears the lowest free bit; free sets the bit of the just-emitted slot, which is by definition occupied). No read-modify-write conflict arises.